The Brand Premium

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Abstract

We highlight the limitations of using cumulative advertising expenses as an input measure of brand value. Using two output measures—Interbrand's data and a novel text-based measure—we find that an equal-weighted portfolio of top brands yields an annual abnormal return of 3%. The excess returns are driven by companies that develop their brands internally. Intangible factors proposed in the literature have no explanatory power for the premium. Analysts underestimate the future earnings of top brands, leading to significant excess returns following earnings announcements. We find no abnormal returns associated with the input measure of brand value. (*JEL* G4, G12, G14, M41)

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A truly great business must have an enduring "moat" that protects excellent returns on invested capital.

The dynamics of capitalism guarantee that competitors will repeatedly assault any business "castle" that is earning high returns. Therefore, a formidable barrier such as […] possessing a powerful world-wide brand (Coca-Cola, Gillette, American Express) is essential for sustained success.

—Warren E. Buffett's 2007 letter to shareholders[1]

A great company is not a great investment if you pay too much for the stock.

—Benjamin Graham[2]

Because almost all investors think these attributes [valued brands and stellar reputations] make for great investments, demand for companies with these characteristics is supersized, which bids up current prices and depresses future returns.

—*Wall Street Journal*[3]

Firm value derives not only from tangible assets but also from intangibles. Intangible assets are generally categorized as knowledge capital (including human capital) and brand capital. Modern corporations are increasingly investing in intangible assets and the role of these assets in explaining firm value has grown significantly over the last few decades.[4] An important intangible asset is brand name, constituting a significant portion of firm value (Belo et al. 2022). While most intangible assets tend to be somewhat invisible, leading to limited attention and undervaluation (Hirshleifer, Hsu, and Li 2013; Cohen, Diether, and Malloy 2013; Edmans 2011), brand names stand out as highly visible and recognized by all market participants. Consequently, unlike typical companies with high intangible assets, brand name companies are susceptible to overvaluation.

The existing literature measures brand capital using the cost-based (or *input*) measure by cumulating advertising expenses and attaching a constant depreciation rate (Vitorino 2014; Belo, Lin, and Vitorino 2014; Belo et al. 2022; Hasan, Taylor, and Richardson 2022). Despite all merits of such measurement,[5] we argue it has limitations that could significantly affect the estimated effect of brand value on an outcome variable as well as the interpretation of that relationship. First, approximately 70% of Compustat firms choose not to report advertising expenses as a separate line item and instead fold them in Selling, General, and Administrative (SG&A) expenses. Importantly, these firms include many top brands, such as Apple, Tesla, Costco, Caterpillar, American Express, General Electric, Booking Holdings, Spotify, and Hilton (for the year 2020). A study of brand value excluding these companies could well reach misleading results. The second issue is that the decision to voluntarily disclose advertising expenses is made strategically (Simpson 2008; Shi, Grewal, and Sridhar 2021). Relatedly, firms strategically cease disclosing advertising expenses. One prominent example is Apple, which suddenly stopped reporting advertising expenses in 2016 (without

[1] See Buffet (2008).
[2] See Graham (2006).
[3] See Moyer (2019).
[4] See Gutierrez and Philippon (2017), Corrado, Hulten, and Sichel (2009), Haskel and Westlake (2017), and Ewens, Peters, and Wang (2024).
[5] For example, to compute a measure of book equity that includes investments in intangible capital, the most appropriate way is to use the cost-based measure.

explanation), a year after its' advertising costs rose 50% to a record $1.8 billion. It was argued that the decision was due to declining Apple's return on advertising investments.[6] Other notable firms with missing advertising expenses in certain periods include Microsoft, IBM, Disney, Cisco, and Starbucks. In summary, most firms choose not to disclose their advertising expenses, and those that do are part of a biased sample created by various trade-offs related to the benefits and costs associated with voluntary disclosure.

The third limitation of the input measure is that it is backward looking and uninformative regarding the quality of the investments. Therefore, companies that suddenly lose a significant part of their brand value (e.g., Motorola) or those that gain a large brand value in a short period of time (e.g., Zoom) will be misclassified based on the input measure. Relatedly, advertising is one of numerous means companies have in hand to improve their brand value, and some companies manage to create a valuable brand while spending nothing on traditional advertising (e.g., Tesla). In fact, marketing scholars have long held the view that "Brands are rarely created by advertising" and that "There is little correlation between the amount spent on advertising and the strength of the brand" (Doyle 1989).

To overcome these limitations, one needs to consider *output* measures of brand value, that is, measures that are forward-looking, reflect the quality (not quantity) of investments in building brands, and are not missing for most of top brands. In this paper, we study two *output* measures of brand value and investigate the relationship between these measures and stock returns. Our first measure comes from Interbrand's yearly list of Best Brands (BB), which is available since 2000.[7] The second (complementary) measure of brand value is a novel text-based measure based on textual analysis of companies' annual reports. Specifically, the measure is based on the simple premise that companies with strong brands speak more about *brand(s)* in their annual reports. For example, while a typical company in 2020 mentions *brand(s)* 3 times in its annual filing, these words appear 210 times in Nike, 119 times in Marriott, and 96 times in Coca-Cola's reports. One main advantage of this text-based measure is that it could be constructed for a large set of companies starting from 1996. Similar to the BB list, we show that this text-based measure also captures many well-known brands (e.g., Tesla, Alphabet, Twitter, Uber, Salesforce, LinkedIn, Grubhub) that are not considered top brands based on the input measure.

Theoretically, whether the stock market fully values brand capital or whether it over- or underestimates it is ambiguous. Scholars have documented the positive effects of brand name on firm value since top brands charge higher prices (Keller 2013; Kaul and Wittink 1995; Ailawadi, Lehmann, and Neslin 2003), have more loyal customers (Chaudhuri and Holbrook 2001), enjoy durable advantages over competitors and face lower threats from entrants (Bronnenberg, Dubé, and Gentzkow 2012), have lower financial constraints and

[6] See O'Reilly (2016).
[7] Interbrand has the most well-known and widely used brand valuation methodology. Section 1.1 provides details about Interbrand and BB data.

higher cash flow stability (Larkin 2013) and elicit greater customer interest in trying, adopting, and personally promoting a new branded product (Keller 1993). Indeed, one of the most widely known bits of advice offered by Warren Buffett is to pick stocks of companies with a sustainable "economic moat," a term that typically refers to companies with strong brands. However, great (strong-brand) companies might not necessarily make great investments. If the stock market is efficient and fully incorporates the value of brands in current stock prices (Fama 1970), top brands should have the same risk-adjusted returns as other companies. In fact, precisely because top brands are great and popular companies, they might be overvalued, leading to lower excess returns for these companies in the future (Frieder and Subrahmanyam 2005).

In contrast, companies with high brand value may be undervalued in the stock market for the following two reasons. First, accounting rules treat investments in creating and enhancing brands as operating expenses, and hence they are fully expensed in the year they are incurred, and no associated (brand) asset is created in the balance sheet. This leads to lower accounting earnings and book value of equity. As a result, a fair price for a company that invests significantly in its brand capital would make it look (misleadingly) overvalued according to common valuation metrics, such as price-to-earnings (PE) or market-to-book (MB) ratios. Second, while it may be easy to observe and incorporate some benefits of a strong brand (e.g., higher margins) into current valuations, many other long-term beneficial effects of a brand name for the firm's value could be underestimated, which is in fact the underlying assumption behind the famous advice of Warren Buffett mentioned earlier.

We find that an equal-weighted portfolio of Best Brands (BB) in the United States earns a monthly excess return of 22 to 30 basis points (bps) from 2000 to 2020. These figures are robust across various factor models and hence not driven by exposures to common factors, such as size, value, momentum, profitability, quality, and expected growth. In addition, the results remain virtually unchanged after controlling for the exposure to intangible-related factors proposed in the literature (Eisfeldt and Papanikolaou 2013; Peters and Taylor 2017; Gulen et al. 2024). Similarly, the results are not due to employee satisfaction (or companies listed in Best Workplaces to Work For), studied in Edmans (2011). Moreover, the excess returns are not driven by micro-cap stocks and as a result a capped value-weighted BB portfolio earns significant abnormal returns. The BB's excess return from Fama-MacBeth regressions controlling for a host of characteristics is 31 bps per month.

To investigate the mechanism behind the excess returns and to further diminish a potential concern that the results we document are driven by other factors correlated with brand value, we conduct two main analyses. First, we exploit the accounting differences in treatment of internally developed versus acquired intangibles. While the former is fully expensed and does not show up in the balance sheet, the latter is capitalized. As a result, companies that develop their brand internally would appear (misleadingly) to be too expensive according to common valuation metrics, such as PE or MB ratios. Indeed, we find that the excess

returns are significant *only* for companies that develop their brands internally. Any alternative hypothesis should also explain why there is no excess return for brand names with a relatively large stock of acquired intangible assets. Second, we study earnings surprises and find that analysts tend to underestimate earnings of BBs in their forecasts. Higher brand value manifests itself in larger positive surprises in future tangible outcomes (i.e., earnings), and we show that a significant portion of the BB's excess return could be attributed to the positive market reaction following earnings announcements.

We find consistent results (with larger magnitudes) when we use our text-based measure of brand value. For example, an equal- and capped value-weighted portfolio of top 100 brands earn a monthly Fama-French 5-factor plus momentum alpha of 57 bps (*t*-statistic, 4.02) and 38 bps (*t*-statistic, 3.02), respectively. This independent analysis using a novel measure of brand value reassures us that the results we document are unlikely to be driven by another factor than the brand name, allows for constructing portfolios with a larger set of brands than what is available via Interbrand, and serves as an out-of-sample test.

The closest paper to ours is Belo, Lin, and Vitorino (2014), in which the authors investigate the relation between brand capital intensity (constructed from past advertising expenditures) and firm risk using a structural model. The general idea in that paper is that companies with a high share of brand capital (to their assets or employees) are low productivity firms since their stock of brand capital is too large. Because investments to build the stock of brand capital (i.e., advertising) is irreversible, firms with high brand capital intensity are riskier and hence have higher expected returns in equilibrium. Our paper complements Belo, Lin, and Vitorino (2014) in two ways. First, our analysis is focused on output (rather than input) measures of brand value. Second and importantly, companies with high brand capital intensity, the focus of the theory in Belo, Lin, and Vitorino (2014), are *not* top brand names. The top 10 companies with highest share of brand capital to assets in 2020 are Everquote, Parts ID, Contextlogic, Zovio, Casper Sleep, Smiledirectclub, Skillz, Cargurus, Curiositystream, and Blue Apron Holding, which are clearly not brand names. In contrast, the top valuable brands in our analysis in 2020 include Apple, Amazon, Microsoft, Coca-Cola, Disney, Nike, Facebook, McDonald's, Intel, and IBM. Hence, our paper is complementary to Belo, Lin, and Vitorino (2014) as we have a different theoretical argument (applied to brand names, not low productivity firms with too much advertising expenses) requiring us to use different measures and study distinct set of companies.

More generally, our paper contributes to the growing literature that studies the asset pricing implications of intangible capital, such as R&D expenditures and patents (Eberhart, Maxwell, and Siddique 2004; Lev and Sougiannis 1996; Hirshleifer, Hsu, and Li 2013; Cohen, Diether, and Malloy 2013; Fitzgerald et al. 2021), selling, general, and administrative expenses (Banker et al. 2019), marketing capability (Ang et al. 2022), trademarks (Hsu et al. 2022), customer satisfaction (Aksoy et al. 2008; Fornell, Morgeson, and Hult 2016), organization capital (Eisfeldt and Papanikolaou 2013), and employee satisfaction (Edmans

2011; Boustanifar and Kang 2022).[8] Most previous papers have studied input measures of intangible capital constructed from SG&A, R&D, or advertising expenses. Our focus on output measures of brand capital is similar to papers that consider employee satisfaction rather than investments on employees (Edmans 2011) or patent citations instead of investments on R&D (Zhen, Lev, and Narin 1999; Hirshleifer, Hsu, and Li 2013; Cohen, Diether, and Malloy 2013).

While we control for extensive set of factors and characteristics, and we provide evidence on the mechanism behind our results, one could think of several alternative potential explanations. First, investments in intangibles might be riskier (e.g., whether advertising leads to higher brand value is uncertain), and therefore these companies could have higher expected returns. This cannot be the explanation since not only do we control for investments on advertising but also our measure of brand value is an output measure identifying companies that do actually have high brand value. The second potential explanation is that companies with high brand value might be those with high organization (human) capital and as argued in Eisfeldt and Papanikolaou (2013), these companies are riskier (with higher expected returns) since their shareholders do not own a significant part of firm assets (i.e., human capital). In contrast, we study the value of brand names, a type of intangible capital owned by the firm. If anything, one would expect that companies with high brand value have lower risk due to their customer loyalty that makes them less vulnerable to macroeconomic shocks, their larger investor base, or their lower crash risk (Cao et al. 2015; Hasan, Taylor, and Richardson 2022). Our regressions also control for organization capital and hence our results are unlikely to be driven by possible correlations between brand value and other intangible capital. Finally, a mispriced omitted and unobserved variable correlated with brand value could potentially be an explanation. To be qualified, such an unobserved variable should also vary systematically across brand names developed internally and those acquired externally. In addition, professional analysts should also systematically underestimate the positive effect of that omitted variable on future earnings. While certainly not impossible to come up with such an unobserved omitted variable and relate it to brand value and excess returns, we find it an unlikely explanation. As we do not have a natural experiment that exogenously assigns different brand values to firms, our paper joins a long list of related studies whose results are subject to noncausal interpretations (Edmans 2011; Hirshleifer, Hsu, and Li 2013; Hsu et al. 2022; Cohen, Diether, and Malloy 2013; Gompers, Ishii, and Metrick 2003; Yermack 2006).

[8] See also Boustanifar and Verriest (2023), who measure intangible liabilities and study its asset pricing implications.

1. Data, Measurement, and Descriptive Statistics

1.1. Best brands list

The main data source in this paper is the annual list of top brands (Best Global Brands) produced by Interbrand, which provides the most well-known and widely used brand valuation method. In fact, the first brand valuation model was developed by John Murphy, the founder of Interbrand, in the 1980s. Murphy realized the need to develop a brand valuation methodology following the wave of mergers and acquisitions in the 1980s. As Murphy himself put it, in the 1980s, there was "a huge buying and selling of branded-goods businesses where what was essentially being bought and sold was brands, but nobody knew how to value brands" (Holdsworth 2001). In 1988, Interbrand developed a proprietary methodology for brand valuation. In 1989, the company evaluated the Pillsbury brand for Grand Metropolitan PLC's acquisition of Pillsbury Co., which is recognized as the first official brand valuation and is considered as a milestone achievement in the branding industry (Arvidsson 2011).

Interbrand's valuation methodology (as will be explained shortly) relies on the idea of measuring incremental future earnings to the branded company that are due to its brand, rather than other firm characteristics, such as product quality. This method of computing brand market value essentially estimates the intrinsic value of a brand name (by discounting its future expected cash flows), which is considered the most reasonable way of estimating the market value of a brand (Damodaran 2009). This is also precisely how marketing scholars define brand equity. For instance Simon and Sullivan (1993) define brand equity as "the incremental cash flows which accrue to branded products over and above the cash flows which would result from the sale of unbranded products."[9] A classic example illustrating brand equity is the significantly higher price of a branded medicine versus a generic version, even though the two products are identical. For example, even for a product like aspirin that has been off patent for more than 100 years, Bronnenberg et al. (2015) find that the price of the branded version produced by Bayer is five times greater than the unbranded products.

Interbrand was also the first company to have its methodology certified as compliant with the requirements of ISO 10668 (requirements for monetary brand valuation). Moreover, its valuation estimates have been found to be relevant and sufficiently reliable for use in financial reporting statements. As a result, Interbrand's (brand) valuation methodology is recognized widely by businesses, standard-setting authorities, academic and regulatory bodies, as well as accountancy and legal practices. Finally and importantly, Interbrand's data on top brands have been available since 2000, which is significantly longer

[9] Prior works have documented a significant brand premium across different categories of products, including automobiles (Sullivan 1998), S&P 500 index funds (Hortaçsu and Syverson 2004), online books (Smith and Brynjolfsson 2001), and consumer packaged foods (Bronnenberg et al. 2015).

than alternatives. A long time series covering different market conditions, that is, booms and crises, is crucial for making conclusions that are reliable and generalizable, particularly since we are studying the long-term stock market valuation of brands. For example, in the dot-com bust of the early 2000s many technology companies with valuable brands were significantly and negatively affected, and therefore an analysis that excludes this crisis may generate biased results.

To be included in the Interbrand assessment, a company must meet a number of criteria. Specifically, the brand needs to be global (i.e., at least 30% of its revenue must come from outside the brand's home region and the company must have a presence in Asia, Europe, and North America), have publicly available data on the brand's financial performance, be visible (the brand must have a public profile and sufficient awareness across major economies of the world), and have the expectation of positive long-term economic profits (the return must be above the brand's cost of capital).

To estimate the value of a brand, Interbrand follows a three-step process. First, it estimates the economic profit of the company or the after-tax operating profit of the brand, minus the cost of capital needed to generate the brand's revenue. In the second step, Interbrand estimates the Role of Brand Index (RBI), which measures the portion of purchases attributable to the brand as opposed to other factors, such as price or product features. Depending on the brand, the RBI is determined by commissioned market research, benchmarking against Role of Brand scores from other brands in the same industry, or expert panel assessment. Multiplying the estimated economic value (estimated in Step 1) by the RBI (computed in Step 2) leads to "brand earnings," which is an estimate of future extra earnings the branded company is expected to earn compared to an otherwise similar company without the brand name. In the third and final steps, brand earnings have been discounted using a measure of brand strength. Brand strength is computed using 10 (internal and external) factors, including leadership, engagement, and relevance, that are all measured relative to other brands in the industry. Overall, brand strength measures the ability of the brand to create loyalty, and therefore sustainable demand and profit into the future.

Starting in 2000, Interbrand began publishing an annual list of BBs. The initial list included 75 global brands in 2000 but it was increased to 100 brands in 2001 onward. Up to 2007, the lists were published in July or August; subsequently, the announcement date was moved to late September or October. As we are interested in whether the stock market overvalues or undervalues brand names, our BB portfolio is formed (or rebalanced) on the first day of the month following the announcement date. The few weeks of delay between the list announcement and our portfolio formation/rebalancing allows sufficient time for the market to react to the news. Therefore, our analysis effectively allows the market to price in the effect of the brand value announcement and then investigates whether the stock market fully incorporates the impact of brand capital on future returns.

1.2. Descriptive statistics of best brands

It is important to note that the published BB list is based on brands, not companies. That is, multiple brands from the same company (e.g., Google and YouTube from Alphabet Inc., Coca-Cola and Sprite from the Coca-Cola Company, and KFC and Pizza Hut from Yum! Brands, Inc.) could show up separately as top brands in the same year. Our analysis is at the stock level and hence we map brands to stocks. This implies that the list of stocks in each year will be lower than the list of brands. Moreover, we focus on U.S. companies for which we have stock price data from the Center for Research in Security Prices (CRSP), which means that private companies are excluded from the sample. Table 1 reports the number of BBs included in our analyses per year. The average yearly number of companies included in the BB portfolio is 47. The number of companies in the portfolio is comparable with that of prior studies that have a similar empirical setup as this paper (Hong and Kacperczyk 2009; Yermack 2006; Edmans 2011). We note that our text-based analysis reported later will have many more firms in the portfolio per year.

Table 1 also reports statistics on the number of companies that are added and dropped from each list compared to the previous one. As expected, the list of BBs tends to be quite stable (i.e., a top brand in 1 year will most likely stay within the top brands the following year), but we do have companies entering and leaving the list. On average, three new brands enter the list each year and two brands leave the list, presumably because of negative shocks to the brand value. For example, Motorola failed to adopt to changes in the telecommunications industry from the wireless telephone to the smartphone and was dropped from the BB list in 2009. Overall, of all companies in the original BB list (2000), 29% had dropped out of the list by 2010 and 45% had dropped out by 2020.

Table 2 reports count of brand names by industry. The number within parentheses following each company name reports how many years (of a total 21 years) each firm was part of the BB list. As shown, some companies have always been part of the list (e.g., Microsoft, Amazon, Apple, HP, McDonald's, Coca-Cola, Nike). In contrast, some other companies that were among the top brands in earlier years had dropped out of the list in later years (e.g., Motorola, Merrill Lynch, Xerox) and some newer brands were added to the list over the past decade that were not part of the list previously (e.g., Salesforce, Netflix).

Figure 1 and Figure 2 present the information reported in Table 2 in a graphic form. Specifically, Figure 1 plots the number of BBs per industry (48 Fama-French classification) in our sample. The figure has two clear messages: (a) BBs have been present in 27 (of 48) industries including Computer Software, Computers, Business Services, Consumer Goods, Hotels, Banking, Electronic Equipment, Machinery, Automobiles, Food Products, Aircraft, and Petroleum and Tobacco Products, and (b) there are relatively large cross-sectional differences in terms of number of BBs per industry. Industries with the most BBs include Computer Software, Computers, Business Services, Consumer Goods, and Restaurants and Hotels, whereas

the ones with the lowest numbers are Recreation, Aircraft, Petroleum and Natural Gas, Insurance, and Personal Services.

Figure 2 illustrates the evolution of industry affiliation of the BBs. The figure shows the number of brands per industry by year. For better visual presentation, the cells are color-coded where the darker cells indicate larger numbers. As shown, there are industries in which the presence of BBs has been stable over time, including Consumer Goods, Restaurants and Hotels, and Banking, whereas industries such as Computer Software and Business Services have been accommodating increasingly more BBs.

Table 3 presents summary statistics of BBs for the first year (in panel A) and the last year (in panel B) of our sample. BBs are clearly large companies with a mean (median) market cap of $288bn ($143bn) in 2020. In comparison, the NYSE 80th, 90th, and 95th percentile breakpoints in 2020 are $17bn, $37bn, and $79bn, respectively. The minimum market value in 2020 is $15.3bn, while the maximum is $2.3tn, hence a very skewed size distribution.

To better illustrate the size distribution of BBs and its evolution over time, Figure 3 plots the percentage of BBs in each NYSE size category per year. As shown, about 90% of BBs have market capitalizations that are larger than the 80th percentile NYSE size breakpoint. We rarely observe a BB with a size below the 50th NYSE breakpoint. Therefore, we present most of our empirical results based on a BB equal-weighted portfolio, since it is unlikely that our equal-weighted results would be driven by micro-cap stocks. We will also investigate the robustness of our results with respect to the exclusion of small stocks and using the value-weighted portfolio. We will show that the straight value-weighted portfolio is dominated by a few extremely large companies and hence we also report the results when we cap the weights of stocks in the portfolio.

Table 3 also show that intangible assets constitute, on average, about 21% of total assets of BBs in 2020, with a large variation ranging from 0% to 54%. It is important to note that intangible assets (e.g., brand values, customer relationships) reported in the balance sheets are entirely those that have been obtained via acquisitions. This is because, except in very specific situations, accounting rules do not allow capitalizing intangible assets that are developed within the firm. In one of our empirical analyses, to shed light on the mechanism behind the return predictability, we exploit this different accounting treatment of internally developed versus externally acquired intangibles.

In addition, as shown in Table 3, BBs spend, on average, 12.01% (7.74%) of their gross profit on advertising in year 2000 (2020). The average ratio masks a significant variation across BBs. There are some BBs that spend relatively little on advertising (e.g., Accenture, Cisco, Deere & Company), whereas some others devote more than 15% of their gross profit to advertising (e.g., Walt Disney, Colgate-Palmolive, Nike). As mentioned earlier, most firms do not report advertising expenses. In 2000 (2020), we have advertising expenses for 28 (38) out of 38 (46) BBs.

Following Eisfeldt and Papanikolaou (2013) and Belo, Lin, and Vitorino (2014), we also compute organization capital and brand capital to total assets for the BBs. As shown in Table 3, in 2020, BBs have an average organization capital to assets of 1, with a huge variation from 0.29 to 2.24. Similarly, the average brand capital to assets is 0.16 with a large variation from almost zero (0.02) to 0.55. These clearly show, not surprisingly as explained above, that top brand names are quite different from each other according to the input measure of brand capital.

Our detailed empirical analysis in the next session investigates whether the BB portfolio is associated with any excess return after controlling for common factors that are known to be related to stock returns.

2. Empirical Methodology and Results

Having documented detailed characteristics and descriptive statistics of the BBs, we turn to statistical analysis to evaluate how much (if anything) of the BB portfolio's return could be considered abnormal. Specifically, in this section, we conduct empirical analyses using time-series factor models as well as Fama-MacBeth regressions to control for characteristics that are known to be related to the cross-section of stock returns. As explained in Section 1.1, the BB portfolio is formed (or rebalanced) on the first day of the month following the announcement date of the BB list. By forming our portfolio in the month after the announcement, the excess returns we report are above and beyond the short-term market reaction to the announcement.[10]

2.1. Time-series factor regressions

The most common practice used in empirical finance to estimate abnormal returns connected with a particular strategy/portfolio is to use factor regressions. Generally speaking, in these setups, we run regressions of excess returns of the portfolio on known (risk) factors that affect returns. A general equation for such a regression is:

$$ER_t = \alpha + \sum \beta_k \cdot Factor_{k,t} + \varepsilon_t, \qquad (1)$$

[10] One might wonder whether the inclusion in the BB list would lead to a positive shock to the discount rate, reducing the stock price and hence increasing the future returns. There is no evidence consistent with this idea. Dutordoir, Verbeeten, and De Beijer (2015) find the opposite effect; the stock price increases following the list inclusion. In our sample, we also find a positive (but statistically insignificant) event return.

where ER_t is the excess returns over the risk-free rate, $Factor_{k,t}$ represents the known factors (depending on the choice of model), and β_k is the factor loading on each long-short factor portfolio return. The coefficient α measures abnormal returns that cannot be associated with exposure of the portfolio to any of the factors.

We use several of the most recent factor models (in addition to Carhart model) to estimate abnormal returns. Specifically, we use the 5-factor model of Fama and French (2018) plus momentum (FF6), q-factor model of Hou, Xue, and Zhang (2014), q5-factor model of Hou et al. (2020), and the "AQR" 6-factor model (Frazzini and Pedersen 2014; Asness, Frazzini, and Pedersen 2019). The FF6 model adds profitability (*RMW*) and investment (*CMA*) to the Carhart model. The q-factor model consists of factors for market (*R_MKT*), size (R_ME), investment (*R_IA*), and profitability (*R_ROE*). The q5-factor model augments the q-factor model with expected investment growth (*R_EG*). Finally, the *AQR* model includes the following factors: market (*MKT*), size (*SMB*), value (*HML*), momentum (*UMD*), betting against beta (*BAB*), and quality minus junk (*QMJ*).

We use all five of the asset pricing models mentioned above (Carhart, FF6, q, q5, and *AQR* models) to test whether the BB portfolio is associated with any significant positive or negative abnormal returns. Unless otherwise stated, the results we report throughout the paper are based on the equal-weighted portfolio, but we do investigate the robustness of these results using alternative methodologies in the next section. The standard errors for coefficients are calculated using the Newey and West (1987) method to allow for residuals to be heteroscedastic and serially correlated.

The results of different factor model regressions are reported in Table 4. We use five different factor models and hence there are five different columns, each corresponding to one model. This table documents a positive and significant excess return associated with the BB portfolio that is robust across all the factor models. For example, the Carhart and FF6 alphas are 30 and 22 bps per month (*t*-statistics, 3.37 and 2.60), respectively. The average estimated alpha across the models (26 bps per month) implies an excess return of 3% per year or a cumulative excess return of 84% during our 21-year sample.

As a robustness check, we also study the excess returns of BB portfolio when we use different benchmark than the risk-free rate in computation of the abnormal returns. Specifically, we compute the returns in excess of industry-adjusted and characteristics (size, value, momentum)-adjusted returns. The results show that the BB portfolio earns a positive and significant annual abnormal return 1.8% (in excess of industry and characteristics-matched portfolios) over a 21-year sample (see Internet Appendix Table 1).

Overall, the results of our analyses in this section suggest that the BB portfolio generates a positive, sizable, and statistically significant excess return, after controlling for common (risk) factors including size, value, momentum, profitability, quality, and expected growth.[11]

2.2. Weighting method

Our main analysis in this paper is focused on an equal-weighted BB portfolio. It is well known that the estimated excess returns could be sensitive to the choice of an equal- versus value-weighted portfolio (Fama and French 2008; Hou, Xue, and Zhang 2020), given the disproportionate effects that "micro" stocks may have on the returns of equal-weighted portfolios. In our setup, however, this should not be a source of concern because, as shown in Figure 3, we rarely have any stock with a market cap below the NYSE 50th size breakpoint. To be more precise, we do not have any BB with a size below the NYSE 30th percentile in any single year during the sample, while there are three occasions (in 2000, 2001, and 2003) where one of the companies in the BB portfolio has a size that lies between the NYSE 30th and 50th percentile breakpoints. Instead, as also highlighted in Fama and French (2008), the characteristics of value-weighted portfolios are dominated by a few very large firms, and this is particularly the case in our context. For example, in 2020, the largest BB has a market capitalization of $2.3tn, compared to the median of "only" $143bn. In other words, the distribution of market capitalization has a few outliers that a value-weighted portfolio is going to be dominated by only a handful of gigantic stocks, which is clearly not a balanced portfolio.

We investigate whether our results are driven by small-cap stocks in two ways. First, we continue to use the equal-weighted BB portfolio but exclude smaller stocks, that is, those with a market cap below the NYSE 50th percentile breakpoint. In the second approach, we create value-weighted portfolios that assign small weights to smaller stocks and hence lower their influence on portfolio returns. Specifically, in the presence of extremely large stocks in the portfolio (as in our situation), to construct a tradable but balanced portfolio one could use capped value-weighted portfolios in which the market caps are winsorized at a threshold, such as the NYSE 80th percentile (Jensen, Kelly, and Pedersen 2023). We follow this practice to create a BB value-weighted portfolio where the market caps for computing weights are winsorized at either the NYSE 80th or 90th percentile breakpoint (to be more conservative). This effectively means that we assume that the BBs with market capitalizations larger than the NYSE 80th (or 90th) percentile breakpoint have market caps equal to the NYSE 80th (or 90th) percentile. In this way, smaller companies

[11] When we replace the standard profitability factor with cash-based operating profitability in FF6, we get a monthly excess return of 28 bps with a *t*-statistic of 2.98. Moreover, adding the liquidity factor to FF6 factors has virtually no effect on the magnitude or statistical significance of the excess returns (these results are not tabulated).

will have small weights in the portfolio and no one stock will dominate the portfolio. We then form our value-weighted portfolio according to these assumptions and run the factor regressions.

Table 5 reports the results, together with the original excess returns we estimated previously for the sake of comparison. As shown and as was expected, excluding small stocks does not have any effect on our estimates of excess returns to the BB equal-weighted portfolio. However, the abnormal returns from the straight value-weighted portfolio are much smaller in magnitude and statistically insignificant. Obviously, the difference between the results of equal- and value-weighted portfolios are not due to inclusion of small stocks, as shown before. On the contrary, the difference comes from the fact that the value-weighted portfolio is dominated by a few very large companies. Indeed, the excess returns to the capped value-weighted portfolios are similar both in size (FF6 alphas of 27 bps and 32 bps for weights capped at 90th and 80th percentiles, respectively) and statistical significance to the ones of the equal-weighted portfolio.[12]

Overall, these results show that the positive and significant abnormal returns to the BB portfolio are not due to the influence of small stocks and robust to using different weighting methodologies.

2.3. Fama-MacBeth regressions

In the previous section, we used factor models to study the abnormal returns of our BB portfolio after controlling for factors from several of the most recent and well-known models in empirical asset pricing. One potential concern related to the factor regression results is that differences in characteristics of BBs (eg., size, market-to-book ratio, profitability, etc.) compared to other firms could be the reason behind the abnormal returns we document. While size, value, and momentum, among other factors, are included as covariates in the factor regressions, these characteristics could still have explanatory power, as argued by Daniel and Titman (1997). In addition, the factor regression set-up does not allow us to control for many characteristics simultaneously.

To address this issue, we use the Fama and MacBeth (1973) methodology (FM). Specifically, we start with the universe of ordinary common stocks in CRSP (share codes of 10 and 11) that are traded in NYSE, Amex, and Nasdaq. We require at least 10 monthly returns per year for a stock to be included in the sample. We then match this data set with accounting data from Compustat. Given that Compustat data are based on fiscal year, we lag annual accounting data by the standard 6 months to make sure they are publicly available

[12] It should also be noted that if the abnormal returns are due to brand value, we should expect higher returns to BB stocks with higher *brand importance*, that is, brand value as the percentage of the market value. Among BBs, however, we observe a negative correlation between brand importance and market value weights. Consistently, in the Fama-MacBeth setting similar as the ones reported in Section 3.3, we find that after controlling for all characteristics (including size, of course), BB stocks with higher brand importance have significantly higher abnormal returns that those with low brand importance.

at that time.[13] We estimate predictive cross-sectional regressions of 1-month-ahead stock returns on the BB indicator and a set of control variables Z_{it} measured at the end of month *t:*

$$r_{it+1} = \alpha_{0t} + \alpha_{1t} BB_{it} + \alpha_{2t} Z_{it} + \varepsilon_{it}, \qquad (2)$$

where r_{it+1} is the return on stock *i* in month *t*+1, BB_{it} is the dummy variable which has a value of 1 if the stock is among the most recent BB list and 0 otherwise. Z_{it} is the set of control variables available at the end of month *t,* and α_1, which is the coefficient of interest, measures the abnormal return associated with the companies in the BB list.

We control for a host of characteristics known to explain cross-section of stock returns. Specifically, we start with the four characteristics corresponding to the Carhart model. Market beta (BETA) is estimated from a regression of monthly excess stock returns on monthly excess market return over a 60-month rolling window. Firm size (SIZE) is measured as the logarithm of the market capitalization and the book-to-market ratio (BM) is measured as of December of the previous year excluding negative book equity firms. Momentum (MOM) is measured as a stock's cumulative return over the past 12 months. We then add the two additional characteristics corresponding to factors of Fama-French 5-factor model together with some further control variables including: ASSET GROWTH is the growth in total assets over the most recent fiscal year, following Cooper, Gulen, and Schill (2008). Operating profitability (PROFIT) is computed as revenue minus cost of goods sold, selling, general and administrative expense, and interest and related expenses, deflated by book equity. Amihud illiquidity (ILLIQ) is computed as the absolute daily return divided by the daily dollar trading volume, averaged over all trading days in a month. R&D is research and developments expenses scaled by market capitalization. Idiosyncratic volatility (IDIO VOL) is defined as the standard deviation of the residuals from a regression of daily excess stock returns on the daily excess market return over a 1-month window. Skewness (SKEW) is the skewness of daily stock returns. All independent variables in the FM regressions are winsorized at the 1st and 99th percentiles.

In further specifications, we add advertising expenses scaled by market capitalization (ADVERT). We also include the stock of brand capital (BRANDCAP) and organization capital (ORGCAP) that are computed using the perpetual inventory method and the same parameters and sample selection as in Belo, Lin, and Vitorino (2014) and Eisfeldt and Papanikolaou (2013), respectively. Both BRANDCAP and ORGCAP are scaled by total assets following the literature. We note that in these additional specifications,

[13] For example, companies whose fiscal year-ends in December file their annual report typically with a few (2–3) months in February or March. We assume that the accounting data for such a firm will be surely available publicly by the end of the following June.

companies with missing advertising costs and brand capital are dropped from the regression, which is the common practice in the literature (Vitorino 2014; Belo, Lin, and Vitorino 2014).[14]

Table 6 reports the results of FM regressions (i.e., average slopes from cross sectional regressions). We use five specifications that gradually add more control variables. In the first specification, we use the Carhart-related characteristics and find that BB stocks have higher future returns of 38 bps per month. The results of the second specification (which should be considered as the full model) show that after adding several additional control variables, the BB abnormal return stays significant at 32 bps per month. In the remaining three specifications, we add ADVERT, BRANCCAP, and ORGCAP, and even though the sample size changes significantly, the BB excess return remains significant and at least as large as what was reported before.

We conduct several additional analyses to study the robustness of our FM results. Specifically, we show that the abnormal returns remain significant when the industry-adjusted returns are used as the dependent variable. We also conduct similar analysis as in Table 6 while keeping observations with missing ADVERT and BRANDCAP. We find significant abnormal returns of between 28 and 31 bps per month in these specifications. Finally, when we replace the BB indicator variable with the logarithm of Interbrand's estimated brand value, we find that, conditional on size and other characteristics, higher brand value is associated with higher future returns. The Internet Appendix reports these results.

Overall, after controlling for a host of characteristics in the Fama-Macbeth setting, we find that BBs have, on average, significant excess returns of around 30 bps per month.

2.4. Time series of excess returns

Having documented significant average excess returns to BBs, one might wonder if the average excess return is driven by a few outlier months during which (for whatever reason) the returns to BBs were extremely large. In addition, it would be interesting to see if the excess returns come from the early part of the sample and have disappeared since. To address these points, we analyze month by month excess returns using FM methodology. We first note that the average excess return is relatively close to the median number. For example, the median monthly excess return is 0.22, which is not far from the average excess return of 0.32 estimated in Table 6, column 2. In addition, we estimate the excess returns in different economic conditions. Specifically, we study the following periods: (a) the dot-com crash from April 2000 to October 2002, (b) the boom leading to the financial crisis from November 2002 to July 2008, (c) the financial crisis from August 2008 to March 2009, (d) postfinancial crisis and the period of sovereign debt crisis from April 2009 to December 2013, and (e) the last period from January 2014 to December 2020. The estimated

[14] In robustness checks, to keep the number of observations constant across specifications, we replace these missing values with zero and add an indicator variable for whether the item was missing. Our results are robust to this treatment.

monthly excess returns in these periods are 43, 11, 158, 37, and 30 bps, respectively. These results are consistent with the idea that brand names do well not only in good times (because of, e.g., stronger demand, having high pricing power, and facing lower threat of entrants) but also in bad times as they typically have more loyal customers.

In our second subsample analysis using FM regressions, we estimate excess returns over 5-year-long periods, namely, 2000–2005, 2006–2010, 2011–2015, and 2016–2020. The corresponding excess returns are 31, 43, 11, and 41 bps, respectively. These results confirm our previous findings that the excess returns are not due to specific events or a particular year during the sample. A month over month estimation of excess returns in the FM setup is shown in Internet Appendix Figure 1, and the cumulative excess return is 196% during the entire sample.

We also plot cumulative monthly alpha plus residuals from different factor models in Figure 4. As shown, there is a clear upward trend with the cumulative excess return of between 67% and 103%, depending on the model.

3. Why Does the BB Portfolio Outperform?

In this section we conduct several additional tests to better understand why the BB portfolio generates significant excess returns.

3.1. Does yearly rebalancing add value?

As shown in Section 1, brand values tend to be persistent, and hence the BB portfolio composition does not dramatically change from one year to another. Having said that, as we also reported earlier, there are reasonable movements into and out of the BB list such that about half of the companies included in the list in 2001, which is the first year with 100 top brands, dropped out of the list by 2020. If the stock market undervalues the value of brand names, we should expect to see a higher return to the BB portfolio that takes into account time-series changes in the BB list versus the one that invests in the original list and never rebalances. Relatedly, investigating this will help us understand whether the excess returns come only from the original set of companies that were included in earlier lists (and remain there) or whether new information about top brands and the development of brand values matters for the portfolio returns.

To investigate this issue, we estimate the excess returns to a portfolio that invests in the 2001 list and never rebalances and compare them with the results we estimated in the previous sections using an annually

rebalanced portfolio.[15] If there is additional value derived from additions and removals from the BB list in subsequent years, we should observe a lower excess return from the portfolio that does not rebalance compared to the original BB portfolio that is rebalanced based on each year's list. Figure 5 shows the excess returns, using all five models, of a portfolio of the 2001 BB list (not rebalanced) together with our originally constructed BB portfolio (with yearly rebalancing) used in Table 4. As we demonstrate, ignoring the stocks added or dropped from the lists in the portfolio construction reduces the excess returns across all five models significantly. For instance, in the Fama-French 6-factor model, taking into account new information (addition/removals of brands) in subsequent lists increases the excess returns twofold from 11 to 22 bps per month. These results strongly suggest that a significant portion of the abnormal returns to the BB portfolio comes from yearly rebalancing of the BB portfolio that removes (includes) the brands that have lost (gained) significant value since the year before.

We also investigate whether there is important information in changes in the magnitude of brand values. Specifically, thus far our investment strategy has been based on the "signal" of the company being listed as a top brand. To understand whether excess returns are associated with stocks with higher dollar brand value, in unreported regressions, we use the FM methodology but instead of a dummy variable (=1 if top brand and =0 if not), we use the logarithm of Interbrand's estimated brand value as our main variable of interest. The estimated coefficient for this variable is also positive and statistically significant, showing that companies with higher brand value also have had higher excess returns.

In summary, our findings strongly support the idea that taking into account updates to the brand values and newer BB lists is important for the magnitude of excess returns we have documented.

3.2. Exposure to intangible factors

As BBs are most likely companies with overall high intangible assets, one might wonder whether exposure of the BB portfolio to intangible (or intangible-related) factors proposed in the literature explains the abnormal returns we document. First, we note that we have already reported abnormal returns using the q5 model, in which the expected growth factor is supposed to capture intangible assets (Hou, Mo, et al. 2020). As shown in Table 4, the BB abnormal return was positive and significant after controlling for the expected growth factor of the q5 model. Moreover, in Fama-MacBeth regressions reported in Table 6, we also included measures of brand capital and organization capital used in the prior literature, none of which has any significant effect on the excess return estimates to the BB portfolio.

[15] We use the 2001 list, since only 75 brand names were announced in the 2000 list while from 2001 onward the list increased to 100 top brands. However, our conclusion remains the same when we use the 2000 list.

Another recent strand of literature has argued that the book value of equity should also include the value of intangibles that are missing from the balance sheet (Lev 2018; Eisfeldt and Papanikolaou 2013; Eisfeldt, Kim, and Papanikolaou 2022; Lev and Srivastava 2020; Park 2022). This literature has also argued that the deteriorating quality of book assets as a fundamental value (because of the omission of intangibles) is at least partially responsible for the underperformance of the value factor during the past decade (Eisfeldt, Kim, and Papanikolaou 2022; Park 2022). In addition, this literature has shown that an HML factor that takes into account intangibles prices assets with lower pricing errors and significantly outperforms the traditional HML factor. Eisfeldt, Kim, and Papanikolaou (2022) also find that an active manager can implement a profitable long-short strategy by going long "intangible" HML and short traditional (i.e., Fama-French) HML. Therefore, one may wonder if the exposure of our portfolio to such an intangible HML could be responsible for the excess returns to the BB portfolio.

To address this point, we replace the value factor in our factor regressions with alternative value factors proposed in the literature that account for intangibles. In particular, we use three such intangible value factors: (1) The HML^{INT} factor constructed by Eisfeldt, Kim, and Papanikolaou (2022) essentially follows the same methodology as the Fama-French HML, with the main difference that it estimates firm-level stocks of intangible assets, as in Eisfeldt and Papanikolaou (2013), and adds these to the book equity of the firm. In this method, intangible assets are measured by applying the perpetual inventory method to flows of all SG&A and R&D expenses. (2) HML^{IME} is an isolated intangible value factor that is computed using sorting of firms based only on their book value of intangible capital. (3) HML^{PTINT} includes only 30% of SG&A plus 100% of R&D as investment in intangibles, following Peters and Taylor (2017).

As shown in the first three columns in Table 7, when we replace the traditional HML factor with these intangible HML factors, the alpha in factor regressions remains virtually unchanged. Moreover, the exposure of BB portfolio to these factors is very small in size and statistically insignificant. These results show that the intangible factors proposed in the literature and the BB portfolio have little correlation with each other.

We add two further robustness checks with respect to intangible capital factors. First, we augment the q5 model with intangible HML factors discussed above. As shown in the columns 4–6 of Table 7, the alpha estimates barely change following this specification. Second, following Gulen et al. (2024), we use FF6 factors that are all adjusted for intangible capital.[16] As shown in column 7 of Table 7, the estimated monthly alpha remains 29 bps (exactly equal to the original estimate).

We also investigate whether the exposure to organization capital factor as in Eisfeldt and Papanikolaou (2013) explains the abnormal returns to BB portfolio. To do so, we augment the FF6 factors with the

[16] We thank the authors for providing us the data on intangible factors.

organization capital factor. We find a negative and statistically significant loading of BB portfolio on this factor. Consequently, the monthly BB excess return increases in both size and statistical significance when we control for organization capital factor.[17] Furthermore, we construct a portfolio which is long (short) on top (bottom) quintile of brand capital companies and use the return on this portfolio as an additional factor along with FF6 factors to investigate whether the exposure to this input measure of brand capital drives the BB excess returns. Again, we find that controlling for this factor has almost no effect on the estimated alpha. These results (which are not tabulated for brevity) are consistent with our earlier arguments that the input and output measures of brand value are quite different.

Overall, our results reject the hypothesis that exposure to intangible-related factors proposed in the literature or the role of other intangible assets (R&D, knowledge capital, etc.) explain the outperformance of the BB portfolio.

3.3. Best brands or best workplaces

Edmans (2011) shows that a portfolio of Best Companies to Work for in America (BC) earns positive abnormal returns using the Carhart model. He shows evidence consistent with undervaluation of employee satisfaction by the stock market. Boustanifar and Kang (2022) find that the main conclusion of Edmans (2011) holds when extending the sample till 2020 and using more recent factor models. Given this, one might wonder whether Best Brand's stocks are also Best Workplaces, in which case the abnormal returns might simply be due to overlap of BB with BC stocks.

We address this point in two ways. First, we run similar factor regressions as before but add the BC portfolio abnormal returns as an additional "factor" in the models. The estimated coefficients for BC returns would show the exposure of the BB to BC portfolio, and the BB alpha would be estimated after controlling for this exposure. Panel A of Table 8 shows the results. Column 1 (3) reports the original BB alpha estimates using Carhart (FF6) model for comparison. Column 2 (4) adds the BC portfolio excess return to Carhart (FF6) model. As shown, the estimated exposure to BC portfolio is not statistically significant. More importantly, controlling for the exposure to BC portfolio barely changes the estimated abnormal returns. For example, the FF6 alpha goes from 22 bps (*t*-statistic, 2.60) to 20 bps (*t*-statistic, 2.35) after controlling for the BC portfolio return.

We also study whether BB abnormal returns are driven by BC stocks in the Fama-MacBeth (FM) setup. Specifically, we run similar FM regressions as before but add an indicator variable which gets the value of 1 for stocks that are listed in the most recent BC list and 0 otherwise. The estimated coefficient for our BB indicator would then show the BB abnormal returns after controlling for BC stocks. Panel B of Table 8

[17] At the firm level, a regression of intangible capital over assets on the BB indicator gives a coefficient estimate of -0.55 (*t*-statistic, -62.6).

shows the results. Column 1 (3) shows the estimated abnormal returns in original FM regressions for the baseline (full) model, reported in column 1 (2) of Table 6. Columns 2 and 4 show the results when we add the BC indicator to our baseline and full models, respectively. As shown, BC gets a positive and significant coefficient, consistent with the results of prior literature that BC firms have higher abnormal returns. However, this inclusion has almost no effect on the estimated BB abnormal returns. For example, in the full model, when we control for the BC effect, the estimated BB coefficient goes from 0.32 (*t*-statistic, 2.27) to 0.29 (*t*-statistic, 2.21).

Overall, our analyses using both factor and FM regressions show that the BB abnormal returns cannot be associated with the correlation of Best Brands' (BB) stocks and Best Companies to Work For (BC).

3.4. Internally versus externally generated intangibles

So far, we have shown that the BB portfolio has a significant and robust excess return, which is not driven by common factors and characteristics associated with the cross section of stock returns. As discussed earlier, one reason for this could be that the balance sheets of companies do not typically reflect their brand values and hence companies with valuable brands would look misleadingly overvalued according to valuation ratios, such as the MB. To investigate this channel directly, we exploit the different accounting treatments of internally developed versus acquired brands.

Generally accepted accounting principles (GAAP) require that expenditures on internally generated intangibles be expensed as incurred while permitting firms to capitalize expenditures on acquired identifiable intangibles, such as brand names.[18] This means that any intangible assets that are listed on a company's balance sheet were most likely gained as part of the acquisition of another business, or they were purchased outright as individual assets. This differential treatment of internally versus externally generated intangibles has been shown to negatively affect the informativeness of reported numbers to external users (Amir and Lev 1996; Lev and Sougiannis 1996). In addition, not showing the value of internally developed intangible assets has been demonstrated to mislead even sophisticated investors and financial analysts, resulting in systematic mispricing of securities for intangible-intensive firms (Aboody and Lev 1998; Eberhart, Maxwell, and Siddique 2004; Lev and Gu 2016).

Therefore, if the lack of information about brand value on the balance sheet causes its mispricing, we should expect lower excess returns to BBs that rely more on building their intangible assets externally (by acquisitions) rather than internally. Marketing scholars have long studied tradeoffs existing in creating versus acquiring brands, leading different companies to prefer different strategies (Doyle 1989; Damoiseau,

[18] An asset is identifiable when it can be separated from the business or when it arises from a contractual or other legal right. For instance, organization capital or customer relationships are not identifiable assets.

Black, and Raggio 2011). Indeed, there is a relatively large heterogeneity in terms of intangible intensity of BBs based on the assets reported on their balance sheets. For example, the ratio of intangible to total assets (as reported on the balance sheet) is more than 50% for companies such as Kraft Heinz, Kellogg's, Pfizer, or Visa, while this ratio is 0 (or close to 0) for companies such as Netflix, Harley-Davidson, Apple, or Nike. Given that all these companies are among top brands, it means that the latter group of companies develops their brands more internally, for which we expect a significantly higher excess return.[19]

To test this hypothesis, we run FM regressions as before, but with two (instead of one) dummy variables: one representing BBs with high intangible intensity as reported in their balance sheet (HIGH INT BB) and the other for those with low intangible intensity (LOW INT BB). HIGH INT BB is a dummy variable and, in each year, gets a value of one for BBs whose intangible assets as the fraction of their total assets are higher than the median and zero otherwise. LOW INT BB are the rest of BBs that their share of intangible assets is lower than the median in each year. The estimated coefficient for HIGH INT BB shows the excess return to BBs that have been acquiring a lot of their intangibles externally. On the other hand, the coefficient for LOW INT BB shows the excess return to companies that build their brand more internally. We expect a lower excess return for HIGH INT BB than for LOW INT BB.

Table 9 reports the results. Columns 1–4 show results for raw stock returns as the dependent variable, whereas columns 5–8 use industry-adjusted returns. There are four specifications for each dependent variable. The first and second specifications correspond to columns 2 (the full model) and 3 (the full model augmented with advertising expenses) of Table 6, respectively. The third specification uses the same model as the second specification but replaces missing advertising data with 0. The fourth specification follows a similar procedure as in the third specification but adds BRANDCAP and ORGCAP as two additional control variables. As shown, the estimated coefficients for HIGH INT BB tend to be positive but much smaller in magnitude than the coefficients for LOW INT BB. For example, as reported in columns 1–4, the excess return to BBs with high intangible intensity is about half of the excess returns to those with low intangible intensity and often statistically insignificant. The difference is even larger in industry-adjusted regressions (columns 4–8), in which case the coefficients for HIGH INT BB drop to below 0.10 with a *t*-statistic of around 0.7. However, the coefficients for LOW INT BB remain sizable and statistically significant, regardless of the specification. This finding that the average positive abnormal return predominantly comes from those BBs that develop their brands internally suggests that lack of reporting of brand values on the balance sheets could be responsible for part of the excess returns we have documented for the BB portfolio.

[19] Our assumption is that the correlation between acquired and internally developed intangible assets is negative. Consistent with this assumption, in a regression of acquired intangibles on internally generated brand capital or organization capital for the sample of BB stocks, we get the coefficients of -0.13 (*t*-statistic, -13.60) and -0.005 (*t*-statistic, -2.22), respectively. Also, these negative correlations exist in the full sample of stocks used in FM regressions, and there is a negative correlation between the acquired intangibles and Interbrand's estimated brand value.

The above-mentioned result is consistent with Chen, Gavious, and Lev (2017), who find that intangible-intense Israeli businesses that are listed in the country and follow IFRS (hence capitalizing the development portion of their R&D) have price informativeness that is an order of magnitude higher than similar Israeli companies that are listed in the United States and follow GAAP (hence expensing R&D completely). Importantly, they also report that before IFRS was adopted in Israel (2007), the share price informativeness of the two groups of firms (Israel-listed and U.S.-listed) was practically identical. Similarly, Oswald, Simpson, and Zarowin (2017) find that R&D capitalization in the United Kingdom has informative value. In addition, Aboody and Lev (1998) exploit the only exception in the United States to the immediate expensing of R&D, that is, capitalization of software development costs, and show that these cumulative software assets reported on the balance sheets are associated with higher stock prices. Moreover, Kallapur and Kwan (2004) find that announcements of brand capitalization (following acquisitions) in a sample of firms with high intangible assets do indeed result in abnormally high (positive) returns.

We note that even for acquired brands, the value that is reported on the balance sheet is the transaction value. It remains unchanged even if the brand value increases significantly over time. This is why even if a company has built its brand completely from acquisitions, the market could still misprice the brand value since the value that is reflected on the balance sheet could be significantly different from the true value. Sinclair and Keller (2017) highlight the importance of providing more information about brand value and its development over time for investors to make more informed decisions.

3.5. Earnings surprises

As discussed in the Introduction, one reason that companies with high brand values could be undervalued is the complexity of considering future benefits of brand names. For example, while it is relatively easy to forecast the future earnings of a well-known brand such as Apple from the sales of its iPhones or iPads, such a well-known brand could easily market a completely new product (e.g., Apple Watch, Apple Car) due to its reputation and the loyalty of customers to its products, something that is much harder to forecast. Valuable brands also have higher market power and could more easily increase the prices of their products, leading to higher earnings. If the market underestimates the future earnings potential of brand names, these companies will earn long-term excess returns, as realized future profits would be larger than expected.

To test this channel, we study earnings surprises. We follow closely the prior literature (Giroud and Mueller 2011; Edmans 2011) and run the following regressions:

$$Earnings\ surprise_{it} = b_0 + b_1 BB_{it} + b_2 Z_{it} + \varepsilon_{it}. \qquad (3)$$

Earnings surprise is measured using 1-year, 2-year, or 5-year surprises. One- and two-year earnings surprises are computed as the actual earnings per share (EPS) minus the median analyst forecast from Institutional Brokers' Estimate System (IBES), deflated by the stock price. Following the prior literature, we also remove observations when the forecast error is large, that is, greater than 10% of the price. A 5-year surprise is the actual 5-year EPS growth minus the median analysts' long-term growth forecast.[20] BB_{it} is an indicator variable for whether the company was included in the most recent BB list. Z_{it} is a set of control variables including size and market to book ratios. We also include year and industry fixed effects in all regressions.

The findings are shown in Table 10. Panels A, B, and C report the results for 1-year, 2-year, and 5-year earnings surprises, respectively. As shown, we find robust evidence across all specifications that earnings surprises are significantly larger for BBs than other firms. We note that these effects are robust even after controlling for size, BM ratio, and year and industry fixed effects. These results are consistent with the idea that even professional analysts tend to underestimate the power of brand names in generating future profits, which effectively contributes to long-term excess returns for investors in these companies.

To test the hypothesis that larger earnings surprises of BBs contribute to their abnormal returns, we study earnings announcement returns. Specifically, we first compute the cumulative abnormal return (CAR) defined as the difference between the firm's compounded stock return and the value-weighted market return over the 3-day window [-1, +1] around the date of the earnings announcement. We then regress CAR on a BB indicator variable. In the full specification, we control for size, book-to-market ratio, and add year and industry fixed effects. Table 11 reports the results, which shows that regardless of the specification, BBs have significantly higher CAR around earnings announcements. In the full specification (column 4), we find that BBs have a higher abnormal return of 0.28% in a 3-day window around the earnings announcements. Assuming four quarterly earnings announcements per year, earnings surprises account for over 40% of the FF6 alpha we have documented. Considering the well-documented phenomenon of post-earnings announcement drift in the literature (Bernard and Thomas 1990), this constitutes a conservative lower bound estimate.

4. A Text-Based Measure of Brand Value

In this section, we construct a novel (and parsimonious) measure of brand value using textual analysis of companies' annual filings. Our approach is based on the simple premise that companies with a strong

[20] To make sure that at the time of the forecast, analysts know the prior earnings, the consensus forecast is taken 4 months after the previous fiscal year-end. Note that almost all 10K reports are filed within 3 months after the fiscal year-end.

brand speak more about "brand" in their annual filings. As an example, while a typical company in 2020 mentions *brand(s)* only 3 times in its annual filing, this word appears 210 times in Nike, 119 times in Marriott, and 96 times in Coca-Cola's annual filings. Such a method could even work for a company like Tesla that is a well-known brand but spends zero on (traditional) advertising. For example, below are some excerpts from Tesla's annual filing in 2010, in which the word *brand(s)* was repeated 45 times:

> "Our intent is to …strengthen Tesla brand, ". "Our principal marketing goals are to … build long-term brand awareness and manage corporate reputation". "To date, we have limited experience with marketing activities as we have relied primarily on the internet, word of mouth and attendance at industry trade shows to promote our brand. To further promote our brand, we may ..." "The strength of our brand has been highlighted by independent authorities. For example, in November 2009, Advertising Age selected us as one of "America's hottest brands" in a special report highlighting the year's 50 top brands."

To construct text-based brand portfolios, we first download the universe of annual filings from the SEC EDGAR platform since 1996, when digitalized filings became mandatory. After removing the duplicates and amendments, we are left with 261,568 filings. We parse each document and count the frequency of *brand(s)* in each filing. we refer to this number as brand frequency. We also count the total number of words in each filing (after removing stop-words such as *a, an, is*), which will be used to normalize brand frequency. Then, we combine this data set with CRSP/Compustat merged data set, resulting in 113,007 firm-year observations with 12,854 unique firms.

We first analyze the output of text-based brand value to check the validity of the proposed measure. Indeed, we find that companies that are known to have valuable brands show up on the top. For example, Coca-Cola, Nike, Marriott, American Express, Gap, McDonald's, Colgate-Palmolive, Altria Group, Starbucks, Mattel, Kellogg, and Bristol-Myers Squibb, among other companies, rank as top brands in almost every year. Other usual suspects, such as Kraft Heinz, Mastercard, Visa, Microsoft, Procter & Gamble, Hilton, Booking Holdings, Raph Lauren, Harley-Davidson, Kellogg, McDonalds, Starbucks, and PayPal, also appear often among top brands. Interestingly, well-known brands but those that are not typically considered as valuable brands according to accounting metrics (e.g., those with little or no advertising expense) also appear in our top brands portfolio. Some examples include Twitter, Alphabet, LinkedIn, Square, Salesforce, Tesla, Pinterest, Uber, Lyft, Snap, and Grubhub.

Companies in this text-based brand portfolio are from diverse industries (39 of 49 Fama-French industries). Only in 2020, the portfolio includes companies from 26 industries including Apparel, Recreation, Food products, Restaurants and Hotels, Beer & Liquor, Retail, Candy & Soda, Consumer

Goods, Computer Software, Entertainment, Business Services, Wholesale, Tobacco products, Machinery and Medical equipment, among others. While much more could be done to properly select a group of top brands, we keep this parsimonious approach, which yields a reasonable outcome. The noise in the data will work against us finding any significant result.

The average (median) brand frequency in the sample is 7.15 (1). The proportion of filings with brand frequency of at least 5, 10, and 20 is 25%, 16%, and 9%, respectively. In 2020, of 3,379 firms, the number (fraction) of firms with brand frequency of at least 5, 10, and 20 is 1,380 (41%), 937 (28%), and 598 (18%), respectively. Therefore, this text-based method allows us to form portfolios of top brands consisting of, for example, 100 or 200 stocks. It will also serve us as an out-of-sample test in the period immediately preceding the Interbrand sample.

Using the above data set, we form portfolios of top (100 and 200) brands based on the normalized brand frequency, that is, brand frequency as a fraction of total words in the filing. We only keep firms with fiscal year ending in December and rebalance portfolios at the end of the following June. To be included in the text-based top brands portfolio in each year, the company should satisfy the following two criteria: (a) it should be ranked among top X (e.g., 100 or 200) companies in terms of normalized brand frequency and (b) it should have a market capitalization of at least $5 bn. The latter condition is to make sure that the portfolio does not include small-cap stocks. As before, we use three weighting schemes: equal-weighted (ew), value-weighted (vw) and value-weighted with weights capped at the 80th NYSE size percentile (vw-capped). The correlation between the ew top-100 text-based and Interbrand-based BB portfolio is 85%.

Table 12 reports the results of time-series factor regressions for the text-based top 100 and 200 brands portfolio (panels A and B, respectively) across all models. In each panel, we report portfolio abnormal returns for the full sample starting from 1996 as well as corresponding alphas for Interbrand sample period reported previously. As shown, we find positive and significant abnormal returns to these text-based brand portfolios which are robust across different factor models and weighting methodologies as before. For example, the vw-capped portfolios of top 100 and 200 firms earn a monthly FF6 alpha of 0.38% (*t*-statistic, 3.02) and 0.25% (*t*-statistic, 2.82), respectively.[21] Moreover, equal-weighted FF6 alpha for the full sample starting from 1996 for top 100 and 200 firms are 0.57% (*t*-statistic, 4.02) and 0.62%. Overall, these abnormal returns are higher than what we reported in Table 4 using Interbrand data. If the stock market undervalues a list of top brands that are publicly announced and receives huge coverage from the media (i.e., Interbrand list), one would expect such undervaluation be higher in companies with significant brand value that receive less attention and coverage.

[21] The corresponding alphas (*t*-statistics) when we use top 150, 250, 300 and 400 brands are 0.26 (2.50), 0.32 (3.43), 0.30 (3.15), and 0.22 (2.63).

Overall, we find consistent and reassuring results using a novel and parsimonious measure of brand value based on textual analysis of companies' annual filings. Specifically, we find that (a) portfolios of top brands based on this measure also earn significant excess returns and (b) these excess returns are larger compared to using publicly announced Interbrand data.

5. Input Measure of Brand Value

The most common method researchers use to estimate the stock of intangible capital is the perpetual inventory method, in which intangible value is computed using cumulative relevant expenses with a constant depreciation rate (Peters and Taylor 2017). This method is applied to estimate the stock of brand capital using advertising expenses (Vitorino 2014; Belo, Lin, and Vitorino 2014). As explained earlier, using the input measure of brand capital, that is, measuring how much companies spend on advertising, could lead to misleading results in our setup for several reasons: (a) 70% of companies and most top brands (e.g., half of top brands in 2020's Interbrand list) do no report advertising expenses, (b) advertising expense is an input measure and uninformative of the quality of the investment in brand, (c) it is backward looking and not responsive to sudden changes in brand value, and (d) valuable brands could be made with almost zero (traditional) advertising (e.g., Tesla), and (e) advertising is only one out of numerous types of expenses that help building a stronger brand. This has long been recognized by marketing scholars as is clear from the following quote from Doyle (1989):

"Brands are rarely created by advertising. This is often misunderstood because advertising is generally much more visible than the factor which creates the differential advantage. For example, Singapore Airlines is a strong brand and does some attractive advertising. But the advertising is not the basis of the brand-rather the advertising communicates and positions it. The basis of the brand is the superior customer service provided by the cabin staff. This, in turn, is largely achieved by Singapore Airlines putting in more cabin staff per place than other airlines. Equally striking is the fact that Britain's strongest brand-Marks and Spencer, has historically done little or no advertising at all. There is little correlation between the amount spent on advertising and the strength of the brand."

Despite these shortcomings, we closely follow prior literature (Belo, Lin, and Vitorino 2014) to choose relevant parameters and measure the firm-level brand capital, and then we study the excess return to portfolios based on this measure. Specifically, we use CRSP/Compustat merged data starting from 1975; before 1975 very few firms were reporting advertising expenses. We also use the depreciation rate of 20% for the stock of brand value. To estimate the initial stock, we use the annual growth rate of 10% for

advertising expenses. Then, we form a portfolio invested each year in companies ranked among top 100 in terms of brand capital in the preceding year and rebalanced at the end of June. While our theoretical arguments for potential undervaluation of brands have no particular prediction related to companies with low level of brand capital (hence our focus in this paper on long-only portfolios), in this section we also show results of a portfolio which is long in the top quintile minus short in the bottom quintile of book value of brand capital to be in line with prior literature that uses the book value of brand capital as a source of risk (a type of investment with adjustment cost) in a structural model (Belo, Lin, and Vitorino 2014).

Table 13 reports the results of time-series factor regressions using both long and long-short equal-weighted portfolios according to the input measure of brand capital. Columns 1–4 show results corresponding to Interbrand period (2000–2020), whereas columns 5–8 report them for the full available sample (1975–2020). As shown, in the long-only portfolio, we find positive and significant Carhart alpha of around 30 bps (columns 1 and 5). However, these excess returns turn out to be only due to the strong exposure of the portfolio to profitability and investment factors. Specifically, after controlling for investment and profitability factors, the excess returns drop significantly in magnitude and become statistically insignificant (columns 2 and 6).[22] In the long-short portfolio, the Carhart alphas are also insignificant and the FF6 alpha becomes negative and even statistically significant in the full sample (-0.39 bps with t-value of -2.03). Our results that there is abnormal returns to the output, but not the input, measure of brand value are somewhat similar as those in Cohen, Diether, and Malloy (2013), who find that that while firms with higher R&D investments do not earn any excess returns, a subsample of firms that are better able to convert R&D to cash flows (based on their past track record) earn abnormal returns of 11% per year.

We also use brand capital intensity, that is, brand capital divided to total assets, as the sorting variable and rerun factor regressions, but we get similar conclusions as reported above. That is, while the Carhart alpha is positive, the FF6 alpha is quite small in size and statistically insignificant (see Internet Appendix Table 4). Similar as in Table 13, the FF6 alpha becomes insignificant due to large, positive, and significant exposure of the portfolio to profitability and investment factors. This itself highlights significant differences across the input- and output-based brand portfolios because the Interbrand portfolio has a very low and insignificant exposure to these factors. Specifically, as reported in Table 4, column 2, the coefficients (t-values) on profitability and investment factors of the Interbrand portfolio are 0.05 (0.78) and 0.01 (0.14), respectively. This is consistent with our arguments before that companies with a large stock of brand capital as a fraction of total assets are quite different from the list of top brands. Importantly, this implies that even if there were a positive premium to stocks with higher (cost-based) brand capital, the economic mechanism

[22] We find similar patterns using value-weighted portfolios. The corresponding alphas (*t*-stat) in columns 1–4 of Table 13 are 0.26 (2.85), 0.11 (1.32), 0.74 (2.72), and 0.01(1.02), respectively.

explaining that premium is unlikely to be the same as the one we proposed in this paper. Instead, a risk-based mechanism coming from large (irreversible) advertising costs, as in Belo, Lin, and Vitorino (2014), could explain the premium to companies with high (cost-based) brand capital.

Overall, we do not find evidence supporting the idea that companies with high brand capital (computed using advertising expenses) earn higher excess returns. This suggests that unlike companies with valuable brands using output and forward-looking metrics, firms with high input measures of brand capital do not appear to be undervalued in the stock market.

6. Conclusion

In this paper, we use two output (as opposed to input and cost-based) measures of brand value and find that portfolios of top brands in the United States earn sizable and statistically significant excess returns. This is, however, not the case when we use the input measure, which we show that it is quite different from the output measures.

Our results are interesting and perhaps somewhat surprising since, unlike other measures of intangible assets that might not be easily observable by the market, brand names are highly visible and as a result are expected to be prone to overvaluation. However, treating of investments on brand equity (not only advertising but also all other efforts to improve reputation and brand value) as an expense as well as difficulty of forecasting long-term benefits of having a recognized brand make companies with valuable brands prone to undervaluation. Consistently, we find that the excess returns almost entirely come from companies that develop their brands more internally (as externally acquired brands are capitalized) and analysts tend to underestimate earnings of companies with valuable brands in their forecasts. Consequently, a significant portion of the excess returns is attributable to the positive market reaction to earnings announcements. Overall, our results are consistent with a mispricing explanation behind the brand premium.

Code Availability: The replication code is available in the Harvard Dataverse at https://doi.org/10.7910/DVN/JTU2GF.

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Figure 1. Counts of best brands' stock, by industry

This figure shows total counts of Best Brands' stocks in each Fama-French 49 industry category during our sample, from September 2000 to December 2020.

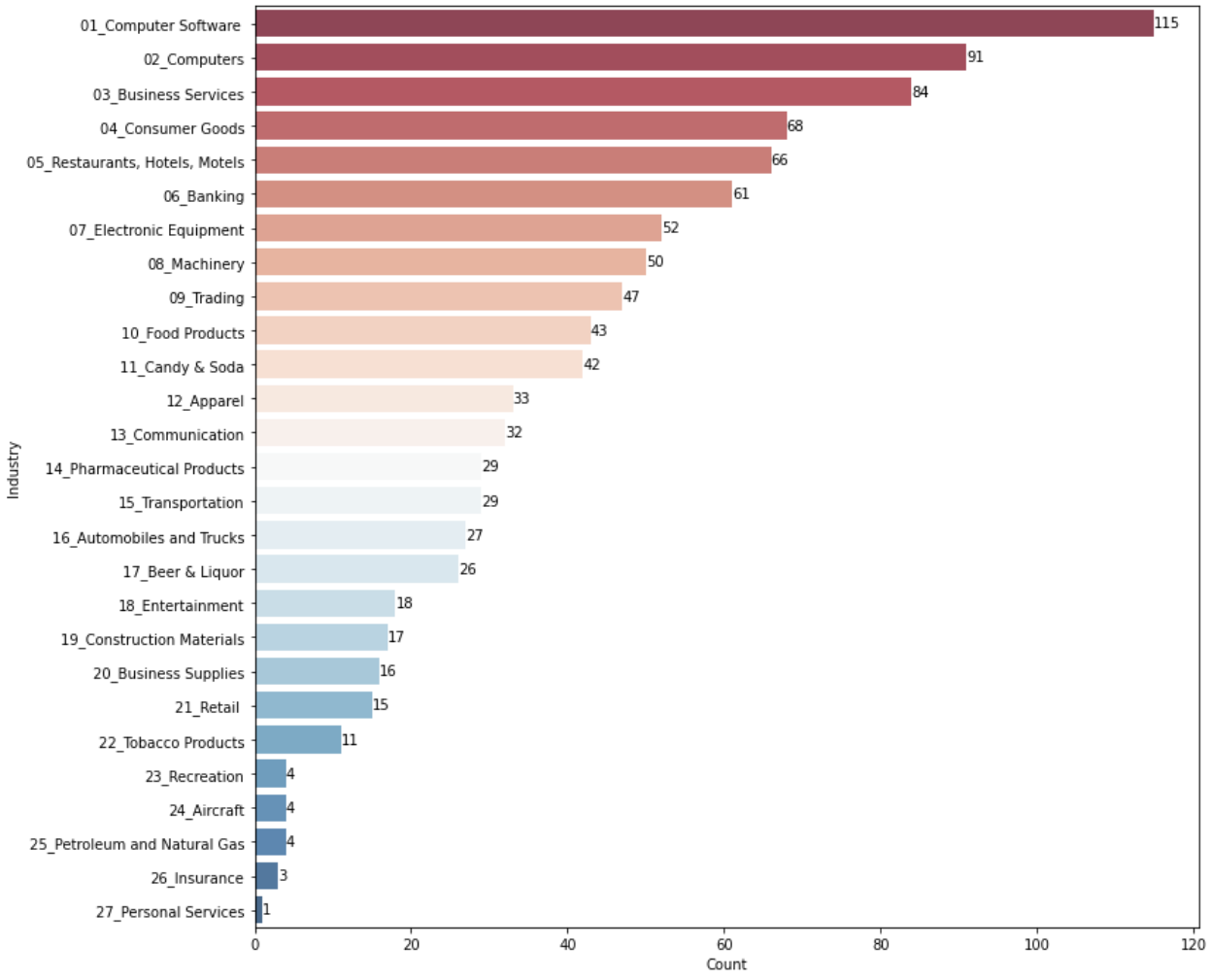


Figure 2. Evolution of best brands' industry affiliations

This figure shows the number of Best Brands' stocks by industry and year. The darker the cell color, the larger the number.

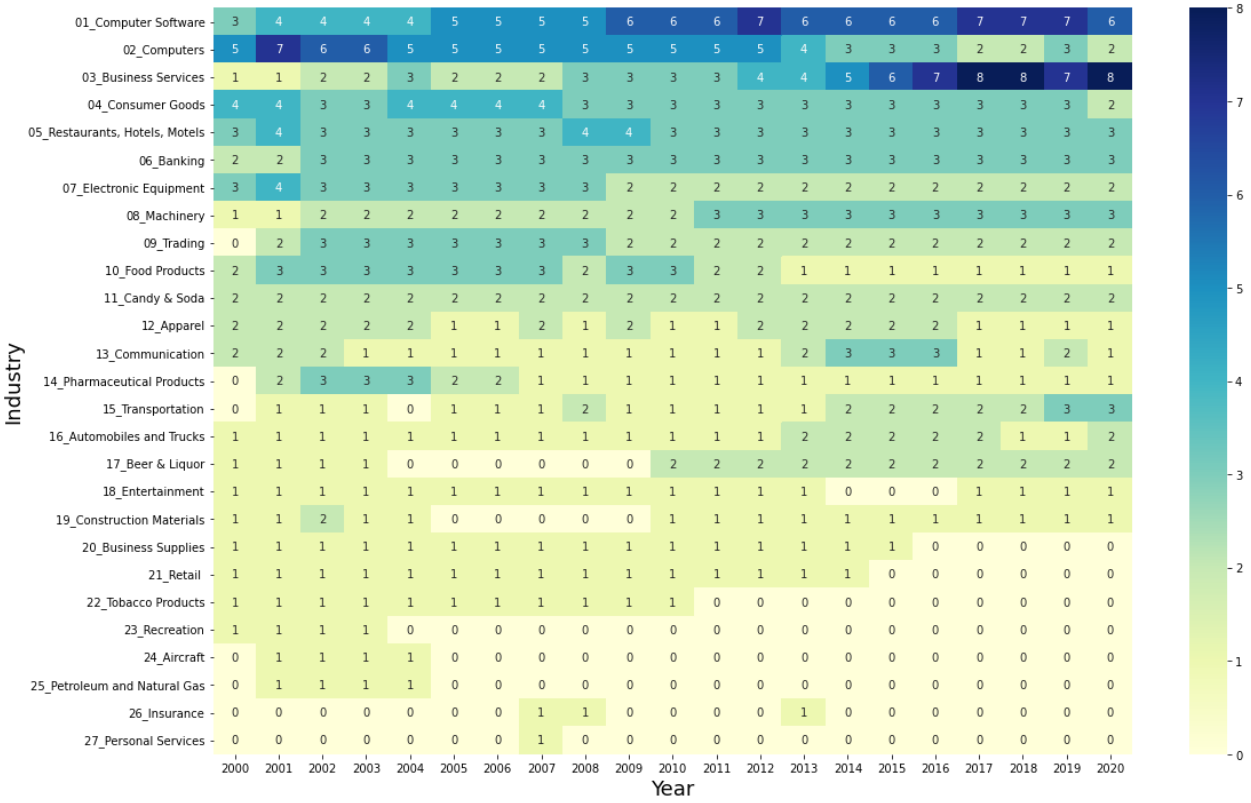


Figure 3. Size distribution of Best Brands' stock across time

This figure plots the percentage of number of Best Brands' firms that belong to each NYSE size category for each year from 2000 to 2020. The categories are between 50th and 80th percentiles (large caps), between 80th and 95th percentiles (mega caps), and above 95th percentile (giga caps). NYSE different percentile breakpoints are taken from Kenneth French's website.

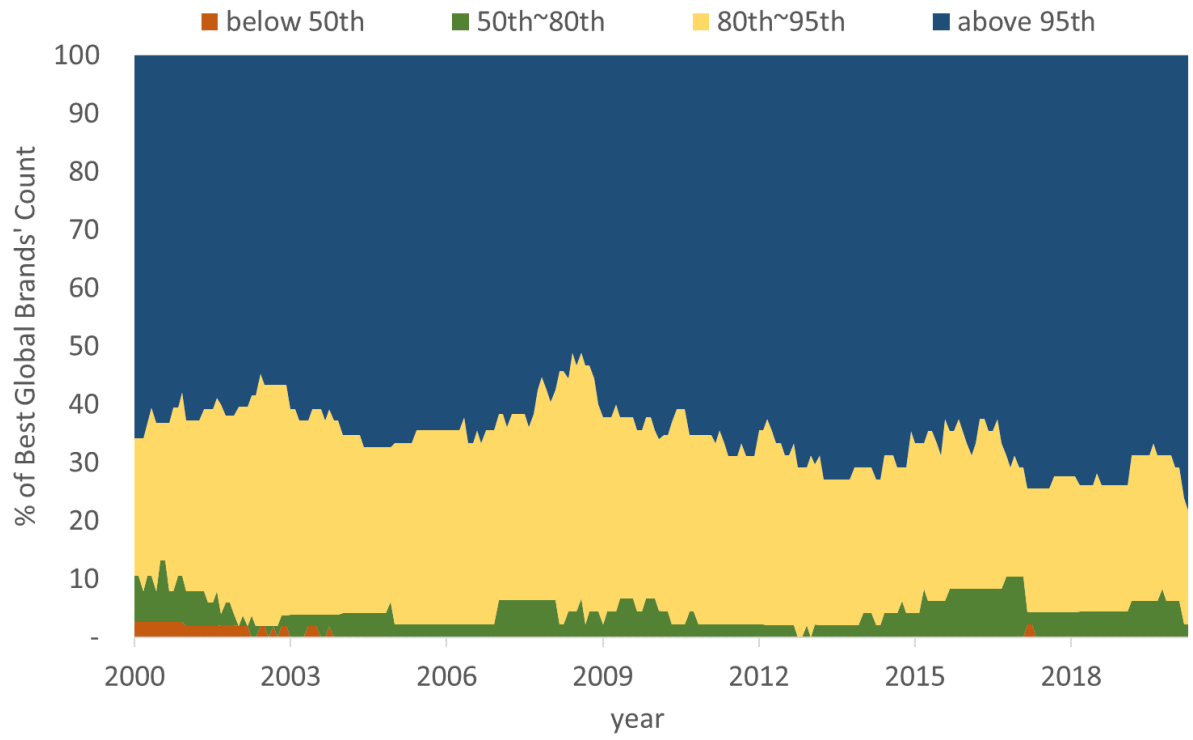


Figure 4. Cumulative excess returns of BBs

This figure plots the cumulative monthly excess returns (i.e., alpha plus residuals) of BB portfolio estimated using different factor models. The sample includes 244 months ranging from September 2000 to December 2020.

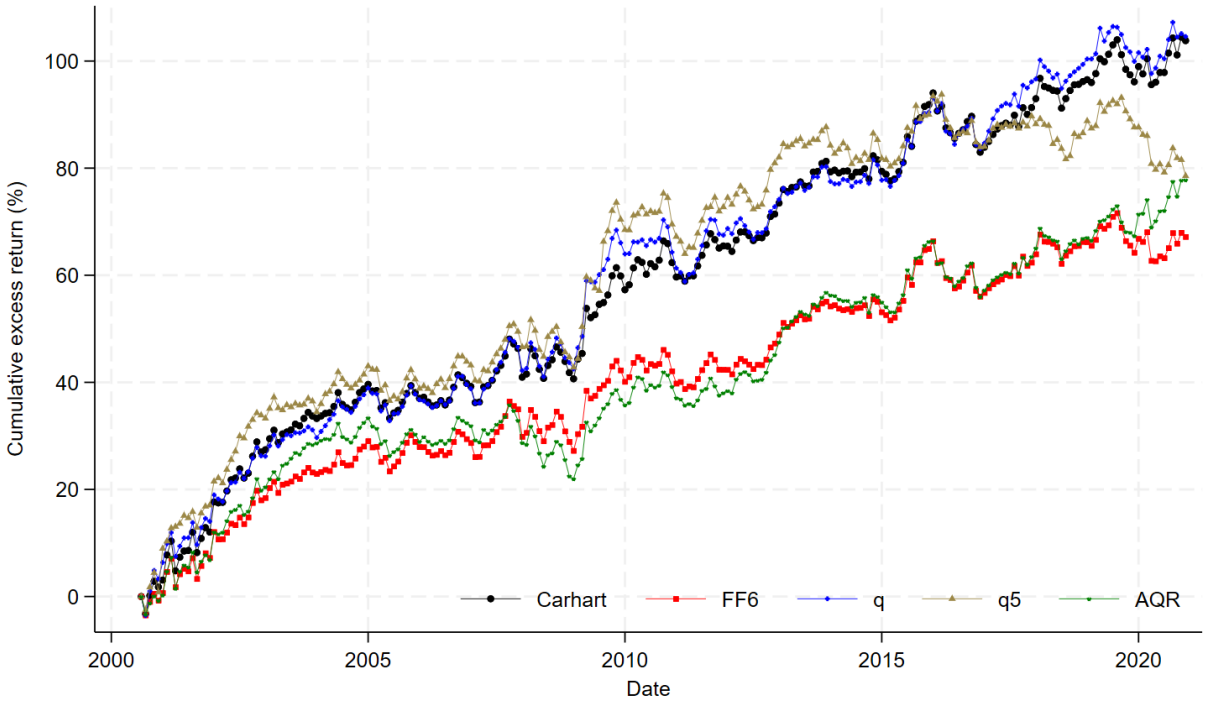


Figure 5. Does yearly portfolio rebalancing add value?

This figure plots the monthly excess returns of the BB portfolio which is rebalanced yearly following the announcement of each Interbrand list of top brands (i.e., BB portfolio). In addition, the figure plots the excess return of a portfolio that invest in the 2001 list of top brands and do not change the portfolio composition during the rest of the sample. We report alphas using five different factor models including Carhart four-factor, Fama-French 5-factor plus momentum (FF6), q, q-5, and AQR 6-factor model. The factors in FF6 are market (Mkt), size (SMB), value (HML), momentum (MOM), profitability (RMW), and investment (CMA). The q-model has market, size (R_ME), investment (R_IA), and profitability (R_ROE), and the q5 model adds to these the expected investment growth factor (R_EG). The AQR model includes the market, size, value, and momentum (UMD), and adds to them betting against beta (BAB) and quality minus junk (QMJ) factors.

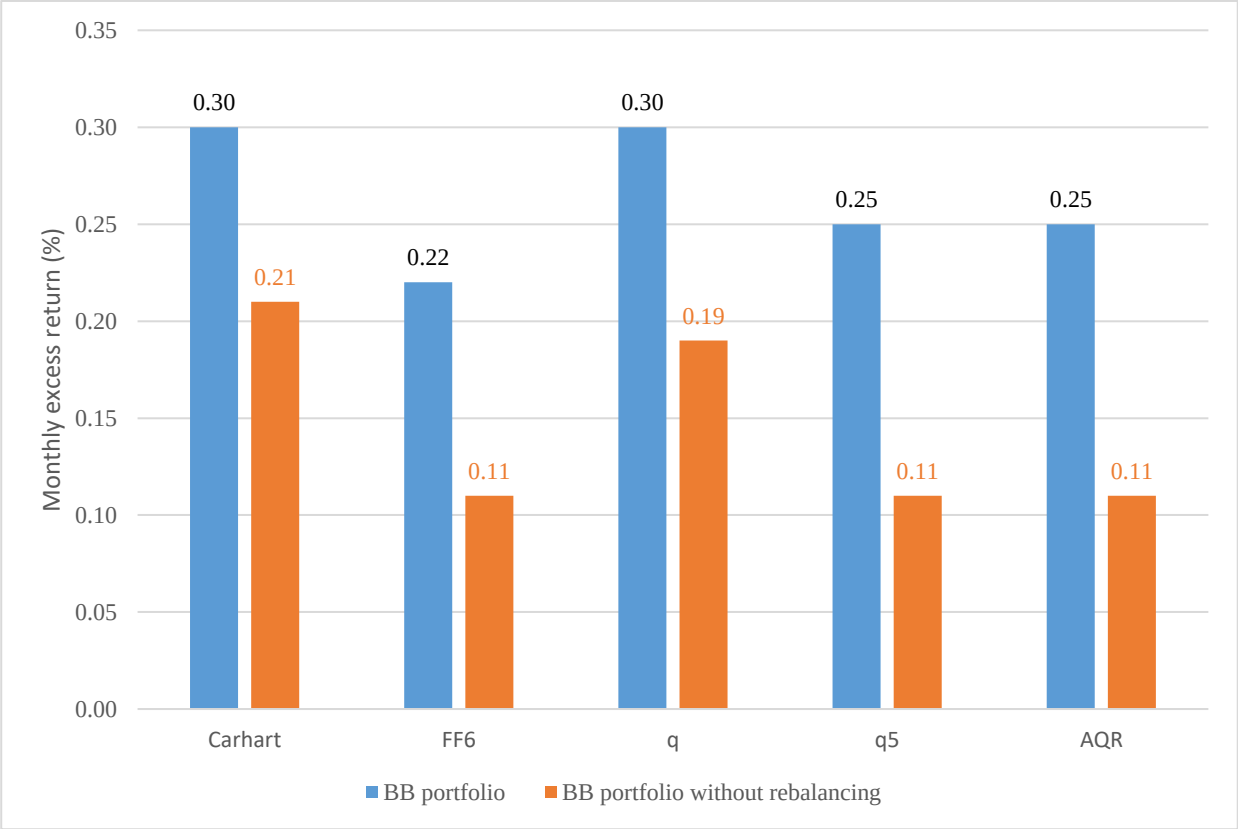


Table 1. Summary statistics on number of BBs used in the paper

This table reports (in column 2) the number of BB stocks per year that are included in the BB portfolio, that is, those that had returns available on CRSP for at least 1 month during the portfolio formation periods of each year. Column 3 shows the number of companies that are added to the list, and column 4 reports the number of added companies as a fraction of total number of companies in the list. Columns 5 and 6 repeat the similar statistics (the raw number and the proportion) for companies that are dropped from the list in each year.

Year of list	**# of BB stocks**	**# added**	**(%)**	**# dropped**	**(%)**
2000	38				
2001	51	13	(25.5)	0	(0.0)
2002	53	6	(11.3)	4	(7.8)
2003	51	0	(0.0)	2	(3.8)
2004	49	2	(4.1)	4	(7.8)
2005	45	3	(6.7)	7	(14.3)
2006	45	1	(2.2)	1	(2.2)
2007	47	3	(6.4)	1	(2.2)
2008	46	4	(8.7)	5	(10.6)
2009	45	4	(8.9)	5	(10.9)
2010	46	3	(6.5)	2	(4.4)
2011	45	1	(2.2)	2	(4.3)
2012	48	3	(6.3)	0	(0.0)
2013	48	3	(6.3)	3	(6.3)
2014	48	1	(2.1)	1	(2.1)
2015	48	1	(2.1)	1	(2.1)
2016	48	2	(4.2)	2	(4.2)
2017	47	2	(4.3)	3	(6.3)
2018	46	0	(0.0)	1	(2.1)
2019	48	2	(4.2)	0	(0.0)
2020	46	2	(4.3)	4	(8.3)
Average	**47**	**3**	**(5.8)**	**2**	**(5.0)**

Table 2. Best brands, by industry

This table reports the brand names counts in each industry group. The numbers in the industry column indicate the total count of stock-years for the best brands in that industry. The numbers in the Brands column show how many years each company was listed as part of the best brands. In cases in which one firm has multiple brands, such as Microsoft/LinkedIn, Google/YouTube, Facebook/Instagram, HP/Compaq, AOL/Time, Gillette/Pampers/Duracell, KFC/Pizza Hut, and Coca-Cola/Sprite, we count the brand as one stock.

Industry	Brands
Computer Software (115)	Amazon.com (21), Microsoft/LinkedIn (21), Oracle (19), Google/YouTube (16), Yahoo! (13), Adobe (12), Facebook/Instagram (9), Salesforce (4)
Computers (91)	Apple (21), HP/Compaq (21), Xerox (16), Dell (14), IBM (14), Sun Microsystems (3), Compaq (2)
Business Services (84)	Accenture (19), eBay (17), Visa (13), MasterCard (9), IBM (7), AOL/Time (5), Hewlett Packard Enterprise (5), PayPal (5), Disney (4)
Consumer Goods (68)	Colgate (21), Harley-Davidson (19), P&G(Gillette/Pampers/Duracell) (18), Kodak (8), Estee Lauder (1), Xerox (1)
Restaurant, Hotels, Motels (66)	McDonald's (21), Starbucks (21), KFC/Pizza Hut (20), Hilton (2), Burger King (1), Marriott International (1)
Banking (61)	American Express (21), Citigroup (21), JPMorgan (19)
Electronic Equipment (52)	Cisco (21), Intel (21), Motorola (9), Texas Instruments (1)
Machinery (50)	GE (21), Caterpillar (19), Deere & Company (10)
Trading (47)	Goldman Sachs (20), Morgan Stanley (19), Merrill Lynch (8)
Food Products (43)	Kellogg's (21), Heinz (13), Kraft (7), Campbell's (2)
Candy & Soda (42)	Coca-Cola/Sprite (21), Pepsi (21)
Apparel (33)	Nike (21), Polo Ralph Lauren (12)
Communication (32)	MTV (17), Discovery (7), AT&T (3), Disney (3), PayPal (1), Zoom (1)
Pharmaceutical Products (29)	Johnson & Johnson (19), Pfizer (6), Merck (4)
Transportation (29)	UPS (16), FedEx (11), UBER (2)
Automobiles and Trucks (27)	Ford (21), Chevrolet (3), Tesla (3)
Beer & Liquor (26)	Jack Daniel's (15), Corona (11)
Entertainment (18)	Disney (14), Netflix (4)
Construction Materials (17)	3M (12), Gillette/Duracell (5)
Business Supplies (16)	Kleenex (16)
Retail (15)	GAP (15)
Tobacco Products (11)	Marlboro (11)
Recreation (4)	Barbie (4)
Aircraft (4)	Boeing (4)
Petroleum and Natural Gas (4)	ExxonMobil (4)
Insurance (3)	AIG (2), Berkshire Hathaway (Heinz) (1)
Personal Services (1)	Hertz (1)

Table 3. Descriptive statistics of BBs

This table shows the descriptive statistics of BBs in the first year (panel A) and the last year (panel B) of our sample. The BB list is announced yearly by Interbrand. Market values are from CRSP, and accounting variables are from Compustat. Organization capital and brand capital are constructed from SG&A and advertising expenses, respectively, and based on perpetual inventory method as in Eisfeldt and Papanikolaou (2013) and Belo, Lin, and Vitorino (2014).

A. Descriptive statistics on BBs' characteristics, 2000

	#firms	Mean	Median	Std.	Min	Max
Market cap ($ bn)	38	84.56	44.41	112.36	0.50	571.61
Market/book	38	13.51	6.17	21.32	0.73	97.43
Dividend yield (%)	38	3.69	1.05	12.22	0.00	94.46
Gross profits to total assets (%)	36	44.28	46.32	21.46	7.65	101.66
Capex to revenue (%)	38	6.95	5.11	4.99	1.34	23.53
Operating cash flows to revenue (%)	38	14.10	12.53	9.41	-3.29	60.74
Intangibles as a % of total assets	38	13.58	7.81	15.60	0.00	75.02
Advertising expense to gross profit (%)	24	12.01	10.02	6.50	4.73	28.77
Organization capital to total assets	20	1.56	1.51	1.02	0.17	3.63
Brand capital to total assets	12	0.33	0.26	0.24	0.11	0.80

B. Descriptive statistics on BBs' characteristics, 2020

	#firms	Mean	Median	Std.	Min	Max
Market Cap ($ bn)	46	288.30	143.09	431.36	14.20	2,255.97
Market/book	43	7.69	5.21	17.40	-26.36	94.55
Dividend yield (%)	45	1.63	1.60	1.33	0.00	5.27
Gross profits to total assets (%)	38	32.54	32.68	13.83	10.48	65.14
Capex to revenue (%)	46	5.50	4.07	3.99	0.00	18.31
Operating cash flows to revenue (%)	45	21.38	20.95	19.48	-38.72	78.43
Intangibles as a % of total assets	46	21.03	16.39	17.68	0.00	54.03
Advertising expense to gross profit (%)	38	7.74	4.95	6.65	0.39	22.61
Organization capital to total assets	35	1.01	0.90	0.57	0.29	2.24
Brand capital to total assets	23	0.16	0.08	0.15	0.02	0.55

Table 4. Excess returns of BB portfolio

This table reports time-series factor regression results of BB equal-weighted portfolio. The dependent variable is the monthly return of BB portfolio in excess of the risk-free rate. We report the results for all five factor models (one per column). The factor models used are Carhart four-factor, Fama-French 5-factor plus momentum (FF6), q, q-5, and AQR 6-factor model. The factors in FF6 are market (Mkt), size (SMB), value (HML), momentum (MOM), profitability (RMW), and investment (CMA). The q-model has market, size (R_ME), investment (R_IA), and profitability (R_ROE), and the q5 model adds to these the expected investment growth factor (R_EG). The AQR model includes the market, size, value, and momentum (UMD), and adds to them betting against beta (BAB) and quality minus junk (QMJ) factors. Newey-West standard errors are calculated, and *t*-statistics are reported in parentheses. The sample is from September 2000 to December 2020. *p < .1; **p < .05; ***p < .01.

Dependent variable:	*Excess return*				
	(1)	(2)	(3)	(4)	(5)
Model:	Carhart	FF6	q	q5	AQR
α	0.30***	0.22**	0.30***	0.25**	0.25**
	(3.37)	(2.60)	(3.18)	(2.11)	(2.59)
β_{MKT}	0.98***	1.01***	1.00***	1.04***	1.05***
	(35.16)	(37.36)	(32.79)	(25.37)	(33.63)
β_{SMB}	-0.02	0.01			-0.01
	(-0.48)	(0.12)			(-0.14)
β_{HML}	0.02	-0.05			0.04
	(0.45)	(-1.12)			(0.74)
β_{MOM}	-0.10**	-0.11***			
	(-2.55)	(-2.91)			
β_{RMW}		0.11*			
		(1.68)			
β_{CMA}		0.11*			
		(1.73)			
β_{R_ME}			-0.05	-0.05	
			(-1.37)	(-1.19)	
β_{R_IA}			0.14**	0.02	
			(2.08)	(0.25)	
β_{R_ROE}			-0.14**	-0.30***	
			(-1.99)	(-3.22)	
β_{R_EG}				0.20**	
				(2.43)	
β_{UMD}					-0.09**
					(-2.38)
β_{BAB}					-0.06
					(-1.64)
β_{QMJ}					0.23***
					(3.63)
# obs	244	244	244	244	244

Table 5. Sensitivity of excess returns of BB portfolio to weighting methods

This table reports the sensitivity of the excess return of equal-weighted BB portfolio with respect to excluding relatively smaller (below NYSE 50th percentile) BBs and value-weighting instead of equal-weighting. We provide excess return estimates for both the straight and capped (at the 90th and 80th NYSE size percentiles) value-weighted results. We report the excess return using five different factor models (one per column). The factor models used are Carhart four-factor, Fama-French 5-factor plus momentum (FF6), q, q-5, and AQR 6-factor model. The factors in FF6 are market (Mkt), size (SMB), value (HML), momentum (MOM), profitability (RMW), and investment (CMA). The q-model has market, size (R_ME), investment (R_IA), and profitability (R_ROE), and the q5 model adds to these the expected investment growth factor (R_EG). The AQR model includes the market, size, value, and momentum (UMD), and adds to them betting against beta (BAB) and quality minus junk (QMJ) factors. Newey-West standard errors are calculated, and *t*-statistics are reported in parentheses. The sample is from September 2000 to December 2020.

Dependent variable:	*Excess return*				
	(1)	(2)	(3)	(4)	(5)
Model:	Carhart	FF6	q	q5	AQR
α original estimates (ew)	0.30	0.22	0.30	0.25	0.25
	(3.37)	(2.60)	(3.18)	(2.11)	(2.59)
α excluding small caps	0.30	0.22	0.30	0.24	0.25
	(3.31)	(3.00)	(2.97)	(2.10)	(2.75)
α straight vw	0.12	0.14	0.16	0.02	0.11
	(1.26)	(1.45)	(1.54)	(0.19)	(1.08)
α capped vw 90th	0.28	0.27	0.33	0.23	0.28
	(3.17)	(3.04)	(2.99)	(2.00)	(2.93)
α capped vw 80th	0.34	0.32	0.38	0.28	0.34
	(3.49)	(3.24)	(3.10)	(2.21)	(3.03)
# obs	244	244	244	244	244

Table 6. Excess returns of the BB portfolio: Fama-MacBeth regressions

This table presents the results (i.e., the average slopes) from predictive cross-sectional Fama-MacBeth regressions. The first specification uses characteristics associated with the Carhart model. The second model is considered the full model with a host of other characteristics for which we have nonmissing values for a large set of observations. The last three models add some extra control variables. Market beta (BETA) is estimated from a regression of monthly excess stock returns on monthly excess market return over a 60-month rolling window. Firm size (SIZE) is measured as the logarithm of the market capitalization and the book-to-market ratio (BM) is measured as of December of the previous year excluding negative book equity firms. Momentum (MOM) is measured as a stock's cumulative return over the past 12-month. We then add the two additional characteristics corresponding to factors of Fama-French 5-factor model together with some further control variables including: ASSET GROWTH is the growth in total assets over the most recent fiscal year, following Cooper, Gulen, and Schill (2008). Operating profitability (PROFIT) is computed as revenue minus cost of goods sold, selling, general and administrative expense, and interest and related expenses, deflated by book equity. Amihud illiquidity (ILLIQ) is computed as the absolute daily return divided by the daily dollar trading volume, averaged over all trading days in a month. R&D is research and developments expenses scaled by market capitalization. Idiosyncratic volatility (IDIO VOL) is defined as the standard deviation of the residuals from a regression of daily excess stock returns on the daily excess market return over a 1-month window. Skewness (SKEW) is the skewness of daily stock returns. ADVERT is advertising expenses scaled by market capitalization. BRANDCAP and ORGCAP are the stock of brand and organization capital (scaled by total assets) that are computed using the perpetual inventory method and the same parameters and sample selection as in Belo, Lin, and Vitorino (2014) and Eisfeldt and Papanikolaou (2013), respectively. All independent variables in the FM regressions are winsorized at the 1st and 99th percentiles. The total number of observations across all regressions and the average r-squared are reported. Newey-West standard errors are calculated, and *t*-statistics are reported in parentheses. The sample is from September 2000 to December 2020. **p* < .1; ***p* < .05; ****p* < .01.

	(1)	(2)	(3)	(4)	(5)
Dep var:			*Stock return (*t+*1)*		
BB	0.38**	0.32**	0.31***	0.50***	0.52***
	(2.44)	(2.27)	(2.73)	(2.64)	(2.87)
BETA	0.05	0.23	0.41**	0.35**	0.35**
	(0.27)	(1.36)	(2.14)	(2.28)	(2.29)
SIZE	-0.06	-0.14***	-0.10**	-0.12*	-0.13***
	(-0.90)	(-3.43)	(-2.40)	(-1.95)	(-2.61)
BM	0.23	0.13	0.16	0.25**	0.25**
	(1.51)	(0.97)	(1.42)	(2.52)	(2.05)
MOM	0.32*	0.14	0.34**	0.45**	0.45**
	(1.69)	(0.71)	(2.40)	(2.36)	(2.37)
ASSET GROWTH		-0.55***	-0.63***	-0.69***	-0.72***
		(-4.90)	(-4.91)	(-2.94)	(-3.49)
PROFIT		0.27*	0.48***	0.67***	0.66***
		(1.77)	(3.45)	(2.86)	(2.62)
ILLIQ		0.02***	0.02***	0.01	0.01
		(3.84)	(3.13)	(1.08)	(0.98)
R&D		-0.11	0.46*	0.23	0.32
		(-0.69)	(1.67)	(0.55)	(0.87)
IDIO VOL		-0.20***	-0.15*	-0.06	-0.05
		(-3.81)	(-1.70)	(-0.44)	(-0.41)
SKEW		-0.09	-0.02	-0.02	-0.02
		(-1.32)	(-0.35)	(-0.27)	(-0.27)
ADVERT			0.18	0.32	0.40
			(0.16)	(0.26)	(0.32)
BRANDCAP				-0.12	-0.07
				(-0.45)	(-0.25)
ORGCAP					-0.01
					(-0.18)
Observations	886,440	857,032	362,914	117,469	117,469
R-squared	.04	.05	.06	.09	.09

Table 7. Excess returns of BB portfolio: Controlling for intangible factors

This table reports the excess returns of equal-weighted BB portfolio when we control for intangible factors. Columns 1–3 replace HML in the Fama-French 5-factor plus momentum (FF6) with three different intangible-adjusted HML factors, namely, HML^{INT}, HML^{IME}, and HML^{PTINT}, respectively. HML^{INT} is an intangible-augmented value factor introduced by Eisfeldt, Kim, and Papanikolaou (2022). This version of HML follows traditional sorting of stocks with the difference that the book equity is the sum of book equity plus the book value of intangibles, measured by applying the perpetual inventory method to flows of SG&A expenses. HML^{IME} is differentiated in terms of firm-sorting method by applying the book value of only intangible assets. HML^{PTINT}, alternatively, includes only 30% of (SG&A minus R&D) plus 100% of R&D as investment in intangible following Peters and Taylor (2017). Columns 4–6 augment the q5-factor model with the three above-mentioned intangible HML factors. Column 7 shows the FF6 alpha when all factors are adjusted for intangibles following Gulen et al. (2024). Newey-West standard errors are calculated, and *t*-statistics are reported in parentheses. The sample is from September 2000 to December 2020. **p* < .1; ***p* < .05; ****p* < .01.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	FF6 w/ INT-adj HML			q5 augmented w/ INT-adj HML			FF6 w/ INT-adj factors
Alpha	0.24***	0.23***	0.23***	0.22**	0.22**	0.24**	0.23***
	(2.87)	(2.75)	(2.76)	(2.08)	(2.05)	(2.09)	(2.60)
HML^{INT}	-0.04			0.15			
	(-0.62)			(1.64)			
HML^{IME}		0.07			0.18*		
		(0.86)			(1.68)		
HML^{PTINT}			-0.03			0.07	
			(-0.70)			(1.42)	
Factor model	FF6	FF6	FF6	q5	q5	q5	FF6
Observations	244	244	244	244	244	244	244

Table 8. Best brands versus best workplaces

This table studies whether the BB excess return is driven by the correlation of Interbrand's Best Brands (BB) list and the list Best Companies to Work For in America (BC). The latter is a yearly list of best workplaces (or places with highest employee satisfaction), studied in Edmans (2011). The first panel our original results of Carhart and Fama-French 5-factor plus momentum (FF6) factor regressions (columns 1 and 3) together with the corresponding abnormal returns when we add the excess return to the BC portfolio as an additional right-hand-side variable (columns 2 and 4). Panel B report our original Fama-MacBeth estimates for the baseline (column 1) and full (column 3) models together with the corresponding results when we add an additional dummy variable indicating whether a company is part of the most recent BC list (columns 2 and 4). Newey-West standard errors are calculated, and *t*-statistics are reported in parentheses. The sample is from September 2000 to December 2020. *$p < .1$; **$p < .05$; ***$p < .01$.

	(1)	(2)	(3)	(4)
A. Factor regressions				
BB alpha	0.30***	0.27***	0.22***	0.20***
	(3.37)	(3.16)	(2.60)	(2.35)
BC		0.13		0.13
		(1.61)		(1.56)
Factor model	Carhart	Carhart	FF6	FF6
B. Fama-MacBeth regressions				
BB	0.38**	0.34**	0.32**	0.29**
	(2.44)	(2.42)	(2.27)	(2.21)
BC		0.20*		0.17*
		(1.94)		(1.96)
Controls	Baseline	Baseline	Full	Full

Table 9. Reported intangibles and excess return

This table repeats the results of Fama-MacBeth regressions as in Table 6 but reports the excess returns separately for Best Brands (BB) that have relatively high intangibles (HIGH INT BB) versus those with relatively low intangibles (LOW INT BB) as reported in their balance sheet. HIGH INT BB is an indicator variable that is 1 if intangible assets reported in the balance sheet over total assets of a BB is above median in the year and 0 otherwise. The dependent variable in columns 1–4 is raw return while it is the industry-adjusted return in columns 5–8. Within each set, the first specification is our full specification corresponding to the second column in Table 6. The second specification (columns 2 and 6) reports results when we add advertising expenses. The third specification within each set (columns 3 and 7) is similar to the second specification with the exception that observations with missing advertising are replaced with 0. The last specification within each set (columns 4 and 8) follows a similar procedure as in the third specification with two additional control variables of brand capital and organization capital. All the control variables' definitions and details of the specifications are reported in the notes to Table 6. Newey-West standard errors are calculated, and *t*-statistics are reported in parentheses. The sample is from September 2000 to December 2020. *$p < .1$; **$p < .05$; ***$p < .01$.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
HIGH INT BB	0.24	0.20*	0.22*	0.17	0.10	0.06	0.09	0.04
	(1.64)	(1.75)	(1.68)	(1.35)	(0.78)	(0.82)	(0.78)	(0.37)
LOW INT BB	0.40**	0.43**	0.38**	0.38**	0.39**	0.46***	0.39***	0.38***
	(2.12)	(2.53)	(2.14)	(2.09)	(2.52)	(3.64)	(2.65)	(2.60)
Main controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Advertising	No	Yes	Yes	Yes	No	Yes	Yes	Yes
Brand/Org cap	No	No	No	Yes	No	No	No	Yes
Require nonmissing additional controls	–	Yes	No	No	–	Yes	No	No
Ind-adjusted returns	No	No	No	No	Yes	Yes	Yes	Yes
Observations	857,032	362,914	857,032	857,032	853,771	361,386	853,771	853,771

Table 10. Earnings surprises

This table reports the results of regressions of earnings surprises on an indicator variable for whether the company was in the most recent Best Brand list (BB), and other control variables (book-to-market ratio and size). The dependent variable in panel A (B) is the actual EPS minus the I/B/E/S median analyst forecast 8 (20) months before the end of the forecast period, scaled by the stock price. In panel C, long-term earnings growth surprise is the actual five-year annualized EPS growth rate minus the median analyst long-term growth forecast. All coefficients are multiplied by 1000. All regressions include year and industry fixed effects as well as a constant. The sample is from September 2000 to December 2020. *$p < .1$; **$p < .05$; ***$p < .01$.

	(1)	(2)	(3)
A. 1-year earnings			
BB	2.19***	2.29***	1.80***
	(3.96)	(3.94)	(2.59)
BM		0.10	0.11
		(0.62)	(0.70)
SIZE			0.48
			(1.62)
# obs	71,090	61,884	61,884
B. 2-year earnings			
BB	3.74***	3.89***	2.09***
	(3.10)	(3.12)	(2.55)
BM		0.35	0.39
		(1.29)	(1.46)
SIZE			1.82***
			(3.25)
# obs	49,755	43,579	43,579
C. Long-term earnings			
BB	3.70***	1.43***	1.31***
	(4.43)	(2.69)	(2.33)
BM		1.65***	1.63***
		(5.63)	(5.57)
SIZE			2.12***
			(3.19)
# obs	13,832	11,848	11,845

Table 11. Earnings announcement returns

This table shows the results of regressions of earnings announcement returns on an indicator variable for whether the company was in the most recent Best Brand (BB) list. The dependent variable is the cumulative abnormal return defined as the difference between the firm's compounded stock return and the value-weighted market return over the 3-day window [-1, +1] around the date of the earnings announcement. Book-to-market ratio and size, both computed at the previous year-end, are used as control variables. The sample is from September 2000 to December 2020. *$p < .1$; **$p < .05$; ***$p < .01$.

	(1)	(2)	(3)	(4)
Dep var:	*CAR around earnings announcements*			
BB	0.25**	0.25**	0.28**	0.28**
	(2.41)	(2.31)	(2.49)	(2.51)
BM				0.12***
				(3.96)
SIZE				0.01
				(0.90)
Year FE	No	Yes	Yes	Yes
Industry FE	No	No	Yes	Yes
Observations	258,621	258,621	258,621	253,110

Table 12. Excess return to a text-based measure of brand value

This table reports time-series factor regression results of BB portfolios constructed using textual analysis of companies' annual filings. The text-based measure of brand value is constructed using the count of words *brand(s)* in each filing divided into the total number of words in the document. To be included in the portfolio in a particular year, a company must meet the following two criteria: (a) ranked among top brands (i.e., top 100 in panel A and top 200 in panel B) in the year according to the text-based measure, and (b) has a market capitalization greater than or equal to $5 bn. The dependent variable is the monthly return of BB portfolio in excess of the risk-free rate. We report alphas for five different factor regressions and for three different weighting schemes: equal-weighted (ew), value-weighted (vw), and capped value-weighted (vw-capped). We also report results (only ew for brevity) using the full sample starting from 1996. Newey-West standard errors are calculated, and *t*-statistics are reported in parentheses. *$p < .1$; **$p < .05$; ***$p < .01$.

	(1) Carhart	(2) FF6	(3) q	(4) q5	(5) AQR
A. Top 100 firms					
ew (1996–)	0.96***	0.57***	0.73***	0.58***	0.54***
	(5.69)	(4.02)	(4.11)	(3.36)	(3.59)
ew (2000–)	1.12***	0.71***	0.99***	0.84***	0.71***
	(6.68)	(5.36)	(6.32)	(5.75)	(5.03)
vw	0.58***	0.21	0.38**	0.28*	0.14
	(3.33)	(1.40)	(2.21)	(1.67)	(0.89)
vw - capped	0.76***	0.38***	0.62***	0.50***	0.34**
	(4.77)	(3.02)	(4.08)	(3.52)	(2.57)
B. Top 200 firms					
ew (1996–)	0.97***	0.62***	0.76***	0.82***	0.63***
	(5.49)	(4.23)	(4.26)	(2.82)	(3.80)
ew (2000–)	1.04***	0.63***	0.92***	0.96***	0.69***
	(5.37)	(4.43)	(4.77)	(3.06)	(4.05)
vw	0.40***	0.12	0.24*	0.21*	0.13
	(2.79)	(1.09)	(1.94)	(1.79)	(1.08)
vw - capped	0.60***	0.25***	0.46***	0.37***	0.27***
	(4.53)	(2.82)	(3.85)	(3.53)	(2.74)

Table 13. Brand capital and excess returns using the input measure

This table reports time-series factor regression results of equal-weighted portfolios sorted according to stock of firm level brand capital. Brand capital is measured using the perpetual inventory method, following Belo, Lin, and Vitorino (2014). The results are reported using two different portfolios: long-only top 100 brands in each year (columns 1, 2, 5, and 6), and a long-short portfolio which is long in stocks ranked on the top quintile of brand capital and short in the bottom quintile (columns 3, 4, 7, and 8). The table also reports the result corresponding to Interbrand sample period (columns 1–4) as well as going back to 1975, the year before which very few firms were reporting advertising expenses. The results for two factor models are reported: Carhart and Fama-French 5-factor plus momentum (FF6). The factors in FF6 are market (Mkt), size (SMB), value (HML), momentum (MOM), profitability (RMW), and investment (CMA). Newey-West standard errors are calculated, and *t*-statistics are reported in parentheses. *$p < .1$; **$p < .05$; ***$p < .01$.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Sample:	Interbrand sample (2000–2020)				1975–2020			
Portfolio:	High *Book* brands		High - Low		High *Book* brands		High - Low	
Model:	Carhart	FF6	Carhart	FF6	Carhart	FF6	Carhart	FF6
α	0.30**	0.12	0.26	-0.24	0.26***	0.12	-0.04	-0.39**
	(2.49)	(1.23)	(0.94)	(-0.95)	(3.08)	(1.40)	(-0.19)	(-2.03)
β_{MKT}	0.97***	1.03***	0.04	0.18***	0.10***	1.03***	0.14***	0.19***
	(27.68)	(39.59)	(0.63)	(3.07)	(35.63)	(39.56)	(2.99)	(4.06)
β_{SMB}	0.26***	0.30***	-0.56***	-0.37***	0.10*	0.17***	-0.80***	-0.55***
	(4.20)	(4.76)	(-3.11)	(-2.96)	(1.69)	(3.97)	(-7.61)	(-6.95)
β_{HML}	0.12*	-0.03	0.58***	0.28***	0.20***	0.07	0.54***	0.34***
	(1.82)	(-0.57)	(5.03)	(2.72)	(3.20)	(1.31)	(3.63)	(3.63)
β_{MOM}	-0.21***	-0.23***	-0.03	-0.11*	-0.19***	-0.21***	-0.03	-0.09
	(-4.75)	(-5.72)	(-0.35)	(-1.97)	(-4.39)	(-5.46)	(-0.31)	(-1.21)
β_{RMW}		0.22***		0.82***		0.27***		0.85***
		(3.23)		(6.95)		(3.81)		(9.64)
β_{CMA}		0.27***		0.22		0.23***		0.15
		(4.33)		(1.26)		(3.81)		(0.97)
# obs	244	244	244	244	534	534	534	534